Econometrica Supplementary Material

SUPPLEMENT TO "TRADE AND THE END OF ANTIQUITY'

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This supplement is organized as follows. Section A describes our data. Section B presents additional empirical results, and section C additional theoretical results. Section D provides extensive information on the coin data assembly, harmonization and cleaning. Section E derives identification results.

A. DATA DESCRIPTION

A.1. *Area of interest*

We define the boundaries of our region of interest in Europe by taking the union of the AD 200 Roman provincial boundaries with the 814 boundaries of the Frankish empire and the area of the West Slavs. The resulting border is roughly the modern-day German-Polish border, plus Bohemia, but follows otherwise roughly the AD 200 Roman Empire boundary. We obtain shapefiles from the *Digital Atlas of the Roman and Medieval Civilizations* (DARMC) hosted at Harvard Unversity (McCormick, Huang, and Gibson, 2007). In the Arab world, the border is delineated roughly by the convex hull of the spatial extent of administrative district boundaries of the extent of the Umayyad caliphate, as given in al-Ṯurayyā (Romanov and Seydi, 2022).

A.2. *Coin Data*

Tables B.I and B.II shows summary statistics for coins and hoards, respectively. See Appendix D for a detailed description of the construction of the coin data.

For the structural analysis, we use the same sample of coins as for the reduced-form analysis in section 2, with the following exceptions: (i) we exclude coins where the mint date interval exceeds 150 years; (ii) we include hoards with a *tpq* up to and including AD 1000 (because we need overlapping generations of coins after our last period to identify our model).

A.3. *Regions*

We define the regions of the ancient world based on political delineations from the 814 political boundaries map of the *Digital Atlas of the Roman and Medieval Civilizations*, and the district boundaries of the caliphate in al-Ṯurayyā. The latter originally contains just district affiliation for each city; we construct regions by assigning space to the district affiliation of its nearest city (Voronoi tessellation).

al-Andalus. Iberian Peninsula, maximum extent of the Umayyad Caliphate 661-750 (DARMC), minus the areas of the Carolingian Kingdom of Aquitaine under Louis in ca. 814 (DARMC).

Aquitaine and Basque Country. Territory of the Iberian Peninsula that is not part of the above definition of *al-Andalus* (i.e. Basque lands), plus the area of the Carolingian Kingdom of Aquitaine under Louis in ca. 814 (DARMC).

Francia and Germania. Kingdom of Charlemagne, ca. 814, plus the area of East Frankish influence (West Slavs), and Brittany (all from DARMC's 814 layer).

Northern Italy and Balkans. Kingdom of Pippin (including Corsica), Papal States, Avar and Croat area, as well as modern-day Bosnia and the Byzantine areas on the Balkans and in modern-day Italy (all from DARMC's 814 layer).
Southern Italy. All regions of modern-day Italy that are not part of *Northern Italy and the Balkans* above, including Sardinia, the Duchy of Benevento, and Sicily.
Byzantine Heartlands. Byzantine territory in modern-day Greece, Turkey (according to DARMC's 814 map), and Cyprus.
al-Sham (Greater Syria). Area of al-Sham (Greater Syria), in al-Ṯurayyā.
Northern Syria and Caucasus. Areas of Aqur (al-Jazirat), al-Rihab (Caucasus), al-Khazar from al-Ṯurayyā.
al-Iraq, al-Jibal, Khuzistan, Kirman. Areas from al-Iraq, al-Jibal, Khuzistan, Kirman, and Fars, from al-Ṯurayyā.
Eastern Caliphate. Areas of al-Mafazat, al-Daylam, Hurasan, Sijistan, al-Sind, and Ma-wara-l-nahr (Transoxiana), all from al-Ṯurayyā.
Jazirat al-Arab and al-Yaman. Areas of Jazirat al-Arab, al-Yaman, and most of the Badiyyat al-Arab, all from al-Ṯurayyā.
Misr (Egypt). Area of Misr, and the area of Barqat (Lybia) east of al-Uqaylah, both from al-Ṯurayyā.
al-Maghrib. Areas of al-Maghrib and the area of Barqat (Lybia) west of al-Uqaylah, both from al-Ṯurayyā.

We use those 13 regions for the structural analysis in sections 4-6. For the reduced form evidence in section 2 (notably for Table I), we use a finer definition of 21 regions: 'Northern Italy and Balkans' is subdivided into 'Corsica & Sardinia,' 'Northern Italy,' and 'Balkans'; 'Southern Italy' is split into 'Benevento & Calabria' and 'Sicily'; the 'Byzantine Heartlands' are subdivided into 'Greece,' 'Thracia & Dacia,' and 'Byzantine Turkey'; 'Northern Syria and Caucasus' is subdivided into 'Aqur (al-Jazirat)' and 'al-Rihab & al-Khazar'; and in North Africa we introduce another region, 'al-Barqat (Lybia)' which in the coarser aggregation is split between Misr (Egypt) and al-Maghrib.

A.4. *Constructing the geospatial model*

We build our geospatial model by combining two geospatial models constructed by historians to model travel distances and routes. The first one, ORBIS (Scheidel, 2015), is a geospatial model of the Roman world and spans roughly the maximum extent of Roman conquests. The second, al-Ṯurayyā (Romanov and Seydi, 2022) is a digitization of Cornu (1983)'s atlas of the Islamic word in the 9th and 10th century. Both geospatial models take the form of undirected graphs; in the case of ORBIS this is augmented by measures of travel costs on each edge. ORBIS also contains sea routes; for the Arab world we augment al-Ṯurayyā with a number of known sea routes. For al-Ṯurayyā we also construct bilateral travel distances by applying the methodology from ORBIS, taking into account the topography of the terrain, and wind directions and speeds for sea routes (Flanders, 2018). Details are available from the authors. We validate these choices by comparing the shortest paths that the model generates to known routes in the Arab world, and by comparing travel times to the ones reported by 10th century Arab geographer Al-Muqaddasī (985) in section B.7 below.

A.5. *Ancient mines data*

Data on ancient mines is from: Stöllner and Weisgerber (2004), Merkel, Oravisjärvi, and Kershaw (2023), Merkel (2016), Morony (2019), Strabo (1917), Rhoby (2019), Mundell Mango (2009), Sarah (2010).

B. ADDITIONAL EMPIRICAL RESULTS

B.1. *Comparison to circulation hoards in Banaji (2016).*

To support the argument that the coins in our hoards are broadly reflective of coin circulation during Late Antiquity, we compare the age distribution of the hoards in our data with a sample of Byzantine circulation hoards described by Banaji (2016), Chapter 6. These are twelve hoards containing between 12 and 751 Byzantine solidi. Figure B.1 shows the average fraction of coins in each 10-year age bin in these hoards, and the distribution of coin ages in Figure 3b, showing a similar age profile. Banaji (2016) also reports that 44% of the coins in these hoards are older than 33 years at time of deposit, compared to 38% in our data (for hoards with more than ten coins).

B.2. *Comparison to the flows of West Roman Terra Sigillata.*

Figure B.2 compares the relationship between distance and coin flows in our data with the relationship between distance and flows of Terra Sigillata in the data of FHLL+ (2021).[1] The distance elasticity is similar but slightly lower for coins, which is potentially due to the fact that naive gravity regressions using coin stocks will exhibit a distance elasticity that is biased towards zero (see Section 3).

B.3. *Within-empire coin redistribution*

One potential explanation for the high coin flows within relative to between empires (political border effect) is that coins could be redistributed across mints before entering circulation, so that the distance from mint to hoard would be less relevant for within-empire flows. Table B.III shows that distance has the same impact of coin flows with and without hoard cell $\times$ empire (that mints the coin) fixed effects in equation (1), suggesting that redistribution within empires was not quantitatively important.

B.4. *Arab conquests and the Mediterranean.*

Figure B.3 shows the number of coins crossing the Mediterranean, and their composition. The Arab conquests (dashed vertical lines) correspond to a decline of north-south flows and an increase in east-west flows, with Islamic coins replacing Roman/Byzantine coins. To further decompose these changes, we estimate by PPML

$$count_{mhpt} = \exp\Big(a_{mh} + a_{mp} + b_1 Mediterranean_{mh} \times After_t + b_2 Mediterranean_{mh} \times After_t \times Islamic_p + u_{mhpt}\Big). \quad \text{(B.1)}$$

We aggregate all hoard (h) and mint (m) locations to $1^\circ \times 1^\circ$ cells, separately for each time period (t), and note for each coin which one of seven aggregate political blocks p had issued it.[2] Count$_{mhpt}$ is the number of coins issued in cell m under empire/dynasty p and found in a cell h, both within time period t. Mediterranean$_{mh}$ is a dummy that is one if the geodetic

[1] Comparing the pairwise flows in the two datasets directly does not make sense since Terra Sigillata are produced in different locations than mints (see Figure 4 in their paper).

[2] These political blocks are: Anglo-Saxon, Sasanian, Roman-Byzantine, Franks, Islamic, Germanic Tribes, and Other Christian. See Appendix Figure D.1 for a breakdown of these and more aggregate political entities.

line between cells m and h intersects the Mediterranean; After$_t$ is a dummy equal to one if t is between 713 and 900, and zero if between 400 and 630; Islamic$_p$ is one if the coin is of Islamic issue (any dynasty); a_{mh} and a_{mp} denote mint cell $\times$ hoard cell and mint-cell $\times$ dynasty/empire fixed effects, respectively. The objective is to investigate whether the Mediterranean acts differentially as a barrier to coin flows after the Arab conquests, and if so, for coins of which issue. Table B.IV presents the results. We drop all mint cell $\times$ empire/dynasty combinations that did not produce coins. Column (1) shows a negative coefficient on the interaction of the Mediterranean and post-conquest dummies, so that after the Arab conquests coin flows declined in cell pairs across the sea. Column (2) shows a positive coefficient on the triple interaction: Islamic coins were facing disproportionately lower barriers on sea routes in the post-conquest world, conditional on origin and destination characteristics. Column (3) contrasts this with Roman/Byzantine coins, which experience disproportionately higher barriers. Column (4) shows similar estimates with hoard cell $\times$ time and mint cell $\times$ time fixed effects, neutralizing potential location-time-specific confounders. All specifications point to the same facts highlighted by Figure 4: there are fewer coins flowing across the Mediterranean in the 8th and 9th century than before; the drop is particularly strong for Roman coins, and the emergence of flows of Islamic coins partly make up for this drop.

B.5. *Coin flows and coin ages*

Figure B.4 uses our data to empirically explore the hypothesis that gravity regressions with flows of durables over longer horizons bias the distance elasticity towards zero. It shows a coefficient plot of the following regression:

$$count_{mth\tau} = \exp\left\{\sum_{\tau' \in T} \beta_{\tau'} \ln distance_{mh} \times 1\left(t-\tau=\tau'\right) + \alpha_{mt} + \alpha_{h\tau} + \varepsilon_{mth\tau}\right\} \quad \text{(B.2)}$$

where $T = \{0, 20, 40, 60, 80, 100\}$ and mint and hoard *tpq* dates are rounded to 20-year intervals. Coins with longer timespans between mint and hoard *tpq* dates are omitted. We estimate the coefficients using PPML.

The results confirm that the distance elasticity for coins that have travelled for longer is lower (i.e. closer to zero) than for coins that have travelled for shorter periods. Section 3.2 and Figure 5 provide the intuition for this result.

B.6. *Estimation of the coin loss rate* (λ)

To estimate λ, we divide coins by their age of deposit (using the *tpq* as the date of deposit) into n-year bins (for $n = 10$ and $n = 20$). We calculate the fraction $f^{(n)}(k)$ of coins in bin $[k, k+n)$, and estimate the parameter of exponential decay from

$$\ln f^{(n)}(k) = \tilde{\lambda}^{(n)} \frac{k}{n} + \varepsilon_k.$$

Table B.V shows the OLS estimation results using 10-year and 20-year bins. λ can be recovered from $\lambda = 1 - \exp(\tilde{\lambda})$, yielding, respectively, $\hat{\lambda}_{10-\text{year}} = 0.15$ and $\hat{\lambda}_{20-\text{year}} = 0.3$.

B.7. *Validating the geospatial model*

We compare the implied travel times from our geospatial model (section A.4) to those reported by the 10th-century Arab geographer al-Maqdisī in his work *The Best Divisions for the*

Knowledge of the Regions (Al-Muqaddasī, 985).[3] Figure B.5 shows the comparison. Our model generates travel times that are slightly larger for shorter distances, and on average similar for longer routes.

B.8. *Statistical tests of random sampling*

Tables B.VI and B.VII present statistical tests of random sampling.

If our random sampling assumption (18) holds, then two random samples of coins drawn from the same region×period should be statistically indistinguishable. Table B.VI present results of such tests for the eight region×period units with the largest number of coins and hoards. We randomly partition the hoards within a region×period unit into two sub-units, and run a statistical test of equality between the probability of a coin being domestic versus foreign in both sub-units, and a test of equality of the age distribution of coins in both sub-units. We repeat the random partition 1,000 times and report average statistics. For all region×period units, coin distributions are statistically indistinguishable.

Further, if the true distributions of coin types differs between two region×period units, our sample of coins between two region×period units should be statistically distinct. Table B.VII present results of statistical tests of equality between all region×period pairs. In 51 out of 56 pairs, coin distributions are statistically different at a 5% level of confidence.

Of course, we recognize that those tests are necessary conditions of random sampling, not sufficient conditions.

B.9. *Alternative accounting method and period lengths*

Table B.VIII shows estimates of the trade cost function parameters for alternative accounting methods and time aggregations. Those robustness checks correspond to changes in data processing. In column 1, we reproduce our baseline estimates, using simple coin counts and 20-year time intervals. In column 2, we use 20-year time intervals, and use only gold and silver coins aggregated according to their relative values (12g of silver for 1g of gold). In column 3, we use counts of coins and 10-year time intervals. In column 4, we use counts of coins and 30-year time intervals.

B.10. *Military conflict and the Justinianic plague*

Table B.IX presents the results from regressions of our baseline estimate for regional real consumption on either the number of battles in a region x period (column 1), or a dummy variable taking value one if at least one battle took place in a region x period (column 2), controlling for period and region fixed effects. Table B.X presents the results from regressions of our baseline estimate for regional real consumption on either the number of plague years in a region x period (columns 1 and 2), or the number of consecutive plague years in a region x period (columns 3 and 4), controlling for period and region fixed effects. Columns 1 and 3 use the full sample, while columns 2 and 4 restrict the sample to the time interval 540-750 during which the Justinianic plague was at its peak.

[3] Al-Maqdisī reports cities and (unsystematically) distances (in travel stages, post stages, and *farsakhs*) or travel times (in days, or nights in the desert) between cities in different parts of the Islamic lands. Historians note that it is unlikely that al-Maqdisī did indeed travel to all these regions, and some distances and travel times are unrealistic. We exclude the most egregious outliers. See the notes of Figure B.5 for details.

B.11. *Tables and figures*

TABLE B.I

SUMMARY STATISTICS: COINS

	count	mean	sd	min	p10	p50	p90	max
Has mint date interval	491,432	0.85	0.36	0	0	1	1	1
Has mint location	491,432	0.55	0.50	0	0	1	1	1
Has mint location and date interval	491,432	0.55	0.50	0	0	1	1	1
Mint date interval, years	416,130	29.63	41.74	0	1	20	58	432
Mint date interval, start year	416,130	464.12	186.58	34	306	375	817	949
Mint date interval, end year	416,130	493.75	184.48	79	333	395	840	950
Age at tpq	416,130	58.99	81.56	0	6	29	154	805
Has material	491,432	0.98	0.15	0	1	1	1	1
Coin is gold	491,432	0.06	0.24	0	0	0	0	1
Coin is silver	491,432	0.17	0.38	0	0	0	1	1
Coin is copper/bronze	491,432	0.74	0.44	0	0	1	1	1
Has denomination	491,432	0.99	0.10	0	1	1	1	1
Has some empire/dynasty information	491,432	0.69	0.46	0	0	1	1	1
Geodesic distance mint to hoard, km	271,051	772.37	783.28	0	68	503	1,631	6,302

Notes: Sample consists of all coins from hoards with tpq between 325 and 950. “Age at tpq” is defined as tpq of the hoard minus the midpoint of the mint date interval.

TABLE B.II

SUMMARY STATISTICS: HOARDS

	count	mean	sd	min	p10	p50	p90	max
Hoard tpq	5,501	590.68	148.21	325	375	578	783	950
Number of coins in hoard	5,501	89.34	823.91	1	1	1	80	43,867
Fraction of coins with mint date interval	5,501	0.98	0.12	0	1	1	1	1
Fraction of coins with mint location	5,501	0.87	0.27	0	0	1	1	1
Fraction of coins with mint date interval and mint location	5,501	0.87	0.27	0	0	1	1	1
Average mint date interval	5,501	23.24	32.66	0	1	11	81	377
Average age of coins at tpq	5,501	25.39	41.76	0	0	10	50	522
Fraction of coins with material	5,501	0.99	0.08	0	1	1	1	1
Fraction of coins that are gold	5,501	0.29	0.45	0	0	0	1	1
Fraction of coins that are silver	5,501	0.15	0.35	0	0	0	1	1
Fraction of coins that are bronze	5,501	0.55	0.49	0	0	1	1	1
Fraction of coins with denomination	5,501	0.99	0.08	0	1	1	1	1
Fraction of coins with empire/dynasty information	5,501	0.80	0.38	0	0	1	1	1
Average distance of coins from mint, km	5,399	683.57	607.55	0	88	534	1,460	6,124

Notes: Sample consists of all coins from hoards with tpq between 325 and 950. “Age at tpq” is defined as tpq of the hoard minus the average coins’ midpoint of the mint date interval.

TABLE B.III

DO COINS GET REDISTRIBUTED WITHIN EMPIRES BEFORE ENTERING CIRCULATION?

	Dependent variable: # Coins_{mdh}			
	(1)	(2)	(3)	(4)
Log Distance	-0.687***	-0.946***	-0.778***	-0.891***
	(0.096)	(0.18)	(0.13)	(0.059)
Empire × Hoard Cell FE	Yes	Yes	Yes	Yes
Mint × Empire Cell FE		Yes		Yes
Sample			Gold only	Gold only
Estimator	PPML	PPML	PPML	PPML
R^2				
Observations	40220	40220	11093	11089

Standard errors in parentheses, clustered at mint cell × empire and hoard cell level.
$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Notes: This table presents variations of equation (1). The dependent variable is the number of coins in a hoard cell h from a mint cell m issued by a political entity p. The regression drops all (m, h) combinations that have no emitted coins. Hoard and mint cells are $1^\circ \times 1^\circ$. Observations only include those that remain after dropping singletons and separated observations. Political entities are categorized into fourteen divisions.

TABLE B.IV

THE MEDITERRANEAN BEFORE AND AFTER THE CONQUESTS

	Dependent variable: Number of Coins			
	(1)	(2)	(3)	(4)
Crossing Mediterranean × After Conquests	-1.947***	-3.288***	-0.696	-1.721
	(0.48)	(0.54)	(0.64)	(1.27)
Crossing Mediterranean × After Conquests × Islamic Coin		7.159***	4.638***	7.416***
		(0.89)	(0.95)	(0.94)
Crossing Mediterranean × After Conquests × Roman Coin			-3.295***	-2.898***
			(0.76)	(0.61)
Mint Cell × Empire FE	Yes	Yes	Yes	Yes
Mint Cell × Hoard Cell FE	Yes	Yes	Yes	Yes
After Conquests FE	Yes	Yes	Yes	
Mint Cell × After Conquests FE				Yes
Hoard Cell × After Conquests FE				Yes
Estimator	PPML	PPML	PPML	PPML
Observations	10430	10430	10430	6180

Standard errors in parentheses, clustered at the hoard × era and mint × era level.
$^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

Notes: This table presents various specifications of equation (B.1). The dependent variable is the number of coins in a hoard cell from a mint cell × dynasty × era (where era is before vs after the conquests). The regression drops all mint × dynasty combinations that have zero emitted coins. Hoard and mint cells are $1^\circ \times 1^\circ$. Flows before the conquests are those with mint date after 400 and *tpq* before 630; flows after the conquests are those with mint date after 713 and *tpq* before 900. Observation counts only include those that remain after dropping singletons and separated observations. "Crossing Mediterranean" is a dummy that is one if the straight line between hoard and mint cell intersects with the Mediterranean. "Islamic Coin" and "Roman Coin" are dummies equal to one if the coin is of Islamic issue (any dynasty) or Roman/Byzantine issue, respectively. "Empires" here are categorized as Sasanian, Roman-Byzantine, Franks, Islamic, Germanic Tribes, and Other Christian.

TABLE B.V

ESTIMATION OF THE COIN LOSS RATE λ

	Dependent variable: Log share of coins in bin $[k, k+n)$	
	(1)	(2)
k/n	-0.161***	-0.354***
	(0.010)	(0.031)
Bin size n	10	20
R^2	0.830	0.817
Observations	55	31

Standard errors in parentheses.

TABLE B.VI

TESTS OF RANDOM SAMPLING WITHIN REGION $\times$ PERIOD

			Test of equal distributions between hoards (p-values)	
	N. coins	N. hoards	Domestic vs foreign coins (Student's t)	Coin ages (Kolmogorov-Smirnov)
al-Sham, Greater Syria (740-760)	13574	124	0.61	0.28
al-Andalus (380-400)	12364	89	1.00	0.69
Francia and Germania (880-900)	7725	25	0.54	0.86
Northern Italy (460-480)	1914	25	0.54	0.43
al-Sham, Greater Syria (680-700)	1625	15	0.61	0.33
Byzantine Heartlands (540-560)	833	109	0.57	0.54
Byzantine Heartlands (680-700)	565	22	0.34	0.59
Southern Italy (640-660)	107	14	0.51	0.12

Notes: We perform statistical tests for the region $\times$ period containing the largest number of coins and hoards. For each region $\times$ period, we randomly partition the sample in two sets of hoards, and aggregate all coins from those hoards within each set. We then run a test of equality between the two sets of the probability of a coin being domestic versus foreign (Student's t test) and of the distribution of coin ages (Kolmogorov-Smirnov test). We repeat this random partition in two sets of hoards 1,000 times, and report the average p-values of those tests. Note that all coins in Iberian hoards (al-Andalus) deposited in the ground in 380-400 are foreign (minted outside Iberia), so that the Student's t test is degenerate and the p-value equal to 1 by construction.

TABLE B.VII

TEST OF RANDOM SAMPLING BETWEEN REGION × PERIODS

A. Test of equal distributions of domestic vs foreign coins (p-values for Student's t test)

	al-Sham Greater Syria (740-760)	al-Andalus (380-400)	Francia and Germania (880-900)	Northern Italy (460-480)	al-Sham Greater Syria (680-700)	Byzantine Heartlands (540-560)	Byzantine Heartlands (680-700)	Southern Italy (640-660)
al-Sham, Greater Syria (740-760)	0.61	0.00	0.10	0.12	0.00	0.00	0.00	0.03
al-Andalus (380-400)		1.00	0.00	0.00	0.00	0.00	0.00	0.00
Francia and Germania (880-900)			0.54	0.85	0.00	0.00	0.00	0.00
Northern Italy (460-480)				0.54	0.00	0.00	0.00	0.00
al-Sham, Greater Syria (680-700)					0.61	0.00	0.00	0.94
Byzantine Heartlands (540-560)						0.57	0.56	0.00
Byzantine Heartlands (680-700)							0.34	0.00
Southern Italy (640-660)								0.51

B. Test of equal distributions of coin ages (p-values for Kolomogorov-Smirnov test)

	al-Sham Greater Syria (740-760)	al-Andalus (380-400)	Francia and Germania (880-900)	Northern Italy (460-480)	al-Sham Greater Syria (680-700)	Byzantine Heartlands (540-560)	Byzantine Heartlands (680-700)	Southern Italy (640-660)
al-Sham, Greater Syria (740-760)	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.08
al-Andalus (380-400)		0.69	0.00	0.00	0.00	0.00	0.00	0.00
Francia and Germania (880-900)			0.86	0.00	0.00	0.00	0.00	0.00
Northern Italy (460-480)				0.43	0.00	0.00	0.01	0.00
al-Sham, Greater Syria (680-700)					0.33	0.00	0.00	0.07
Byzantine Heartlands (540-560)						0.54	0.00	0.00
Byzantine Heartlands (680-700)							0.59	0.04
Southern Italy (640-660)								0.12

Notes: Off-diagonal elements—Panel A reports the p-value of a Student's t test of equality of the distribution of domestic versus foreign coins between two region × period (row and column). Panel B reports the p-value of a Kolmogorov-Smirnov test of equality of the distribution of coin ages between two region × period (row and column). Diagonal elements—same tests within each region × period (randomly partitioned in two sets of hoards, 1,000 times, averaged)—replicate the last two columns of table B.VI.

TABLE B.VIII

DETERMINANTS OF ANCIENT TRADE COSTS: ROBUSTNESS

	Log Trade Costs			
	(1)	(2)	(3)	(4)
Log Travel Time	-3.17	-0.97	-3.5	-3.0
	(0.02)	(0.03)	(0.03)	(0.02)
Political Border	-0.54	-3.44	-2.97	-1.19
	(0.02)	(0.05)	(0.04)	(0.02)
Religious Border: East	-4.28	-4.08	-4.83	-4.25
	(0.22)	(0.59)	(0.29)	(0.28)
Religious Border: West	-1.84	0.23	0.37	-1.21
	(0.12)	(0.31)	(0.26)	(0.12)
Religious Border: Mediterranean	-5.0	-2.3	-3.46	-4.06
	(0.19)	(0.18)	(0.2)	(0.28)
Time Interval	20 years	20 years	10 years	30 years
Coin Accounting	Number	Value	Number	Number
Sample	All	Gold/Silver	All	All
Observations	4,403	2,018	7,412	3,238

Notes: The table shows alternative specifications for the estimates of the trade cost parameters. Column 1 reproduces our baseline estimates, 20-year time intervals and the parameterization of trade costs in column 2 of table III. Column 2 uses 20-year time intervals and weights coins by value (in equivalent grams of gold, assuming a constant exchange rate of 12g of silver for 1g of gold; and using gold and silver coins only). Columns 3 and 4 show variations of the baseline specification where time periods are aggregated to 10-year intervals (column 3) or 30-year time intervals (column 4), adjusting the coin loss rate to the period length (always 1.7% p.a.).

TABLE B.IX

MILITARY CONFLICT AND REAL CONSUMPTION

	Dep. var.: Log Real Consumption X_n/p_n	
	(1)	(2)
$(\text{\# Battles})_{it}$	0.159	
	(0.13)	
$(\text{At least one battle})_{it}$		0.757*
		(0.42)
20-year time period FE	Yes	Yes
Region FE	Yes	Yes
R^2	0.368	0.371
Observations	403	403

Standard errors in parentheses, clustered at the 20-year time interval level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE B.X

PLAGUE EVENTS AND REAL CONSUMPTION

	Dep. var.: Log Real Consumption X_n/p_n			
	(1)	(2)	(3)	(4)
(# Plague years)$_{it}$	0.555*	-0.0585		
	(0.30)	(0.23)		
(# Consecutive plague years)$_{it}$			0.552	0.307
			(0.86)	(0.56)
20-year time period FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
Sample	Full	$[540, 750)$	Full	$[540, 750)$
R^2	0.371	0.742	0.366	0.742
Observations	403	143	403	143

Standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is our estimate of $\log X_n/p_n$. The independent variables are the number of plague-years, and the number of consecutive plague-years in a 20-year time interval and region, from Stathakopoulos (2004). Columns 2 and 4 restrict the sample to the period $[540, 750)$, which is the period from the onset of the Justinianic Plague to the end of period covered by Stathakopoulos (2004).

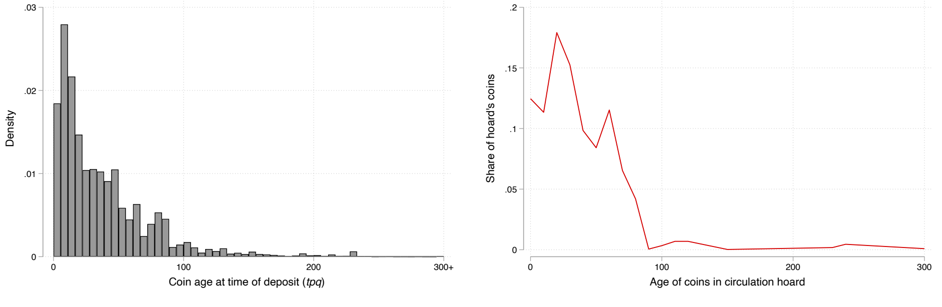


FIGURE B.1.—Comparison with Circulation Hoards in Banaji (2016)

Notes: The left panel reproduces Figure 3b. The right panel shows the average share of coins in each 10-year age bin in the circulation hoards of Banaji (2016), Chapter 6, who reports the issuing emperors (but not mint dates) of the coins in these hoards. We draw mint dates uniformly from the ruling years of these emperors.

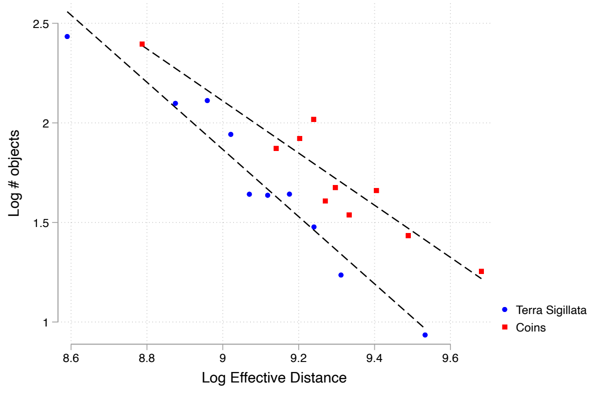


FIGURE B.2.—Comparison with flows of West Roman Terra Sigillata (FHLL+, 2022)
Notes: The figure shows a binscatter of the log number of objects flowing (either coins or number of Terra Sigillata) between two 0.5 × 0.5 degree cells, against the log effective distance between cells. Both are de-meaned by origin and destination location. Cell definitions and effective distances are from FHLL+ (2021). The coin data is restricted to hoards with *tpq* up to 450 and that lie within the aforementioned cells.

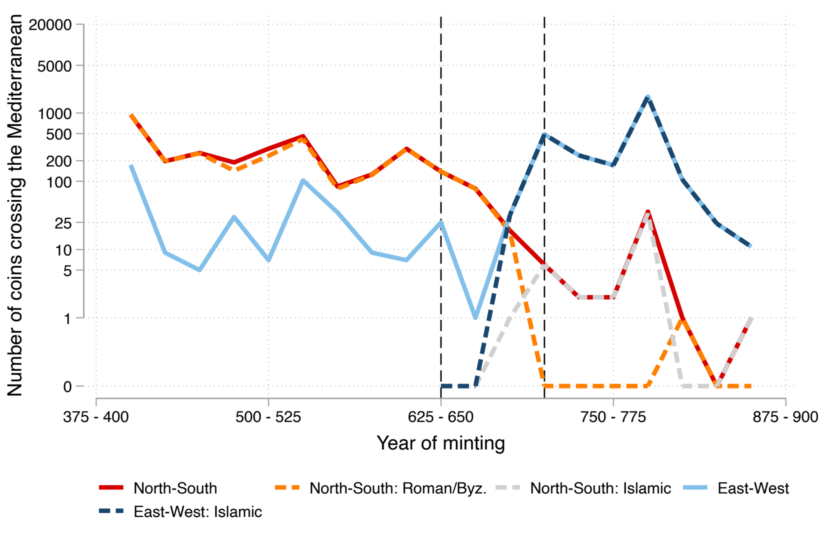


FIGURE B.3.—Number of coins flowing across the Mediterranean

Notes: The figure shows the number of coins minted in the 25-year interval on the horizontal axis, that are minted on one side of the Mediterranean, and are found on the other. Flows include both directions. The north is defined to go from the Pyrenees to Byzantine Turkey, east from Byzantine Turkey to Egypt, south from Egypt to the Maghreb, and west from Maghreb to Aquitaine. Border regions are included in these definitions, so regions are partly overlapping. The vertical lines correspond to the Arab conquest, starting in 622 under Muhammad and ending in 713 with the conquest of Iberia.

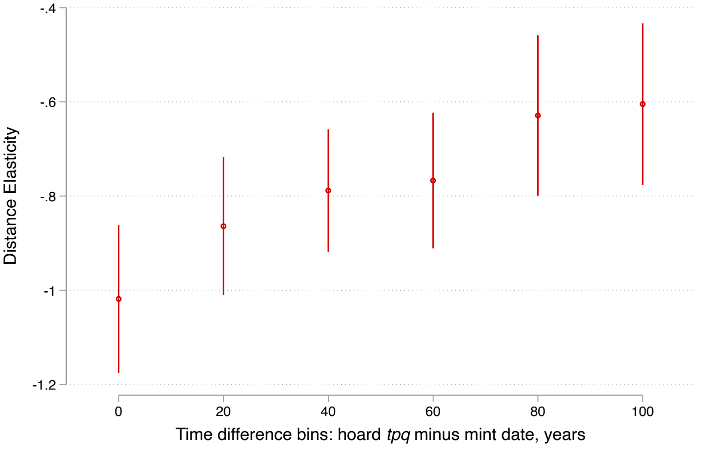


FIGURE B.4.—The distance elasticity declines as coins age

Notes: The figure shows the distance elasticity estimates when estimating equation (B.2) using PPML.

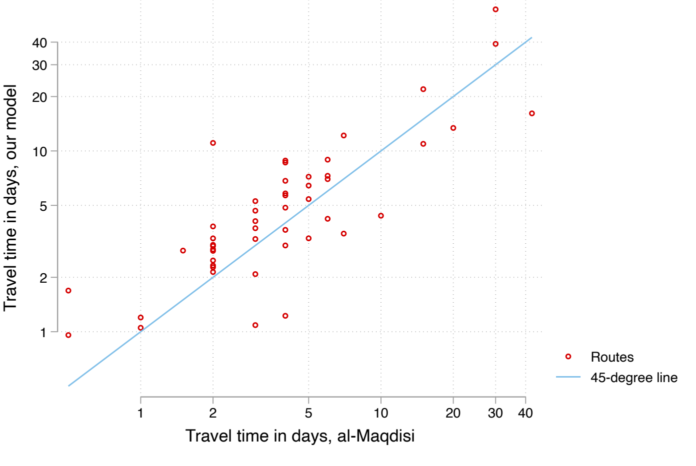
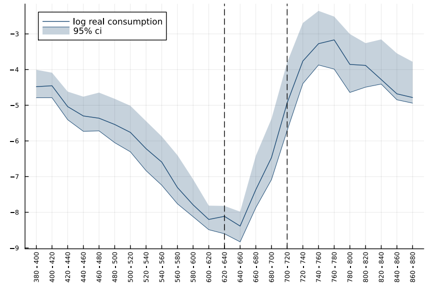
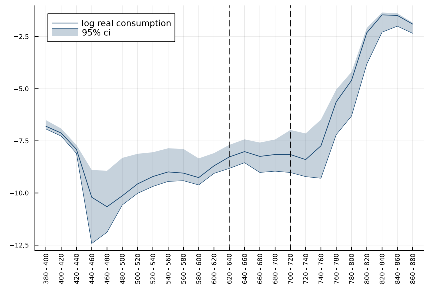


FIGURE B.5.—Comparing travel times to Al-Muqaddasī (985)

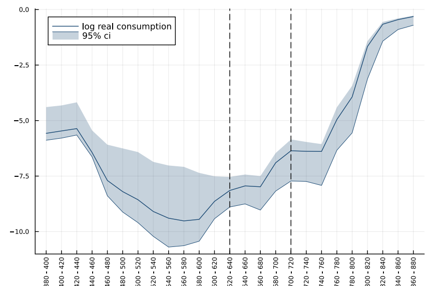
Notes: Each dot refers to a city-to-city connection for which Al-Muqaddasī (985) lists the travel time in days. We exclude desert routes (where he reports travel times in nights or in watering stations, routes with cities that cannot be found in al-Ṯurayyā, as well as the route between Tahart and Fes, which he claims can be travelled in three days despite it being a distance of more than 600 kilometers (our model predicts 22 days).



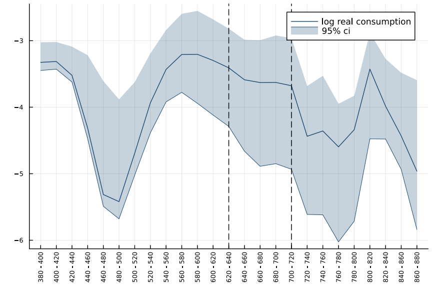
(a) Al-andalus



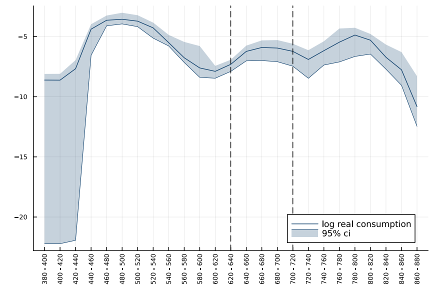
(b) Aquitaine Basque Country



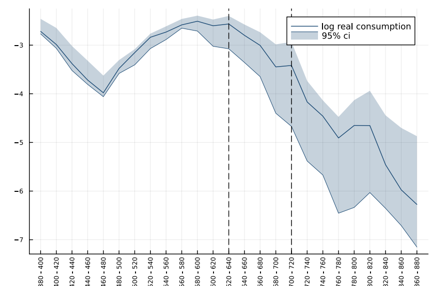
(c) Francia and Germania



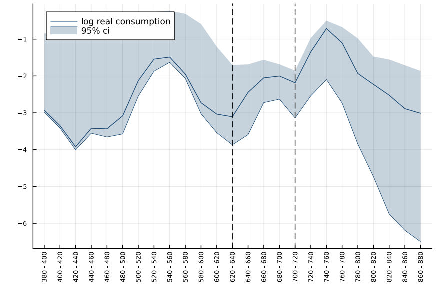
(d) Northern Italy and Balkans



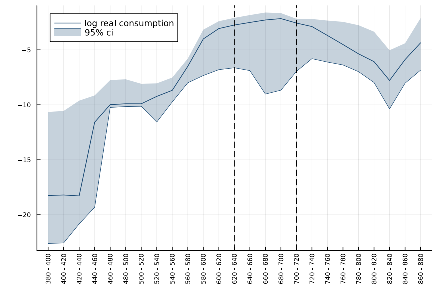
(e) Southern Italy



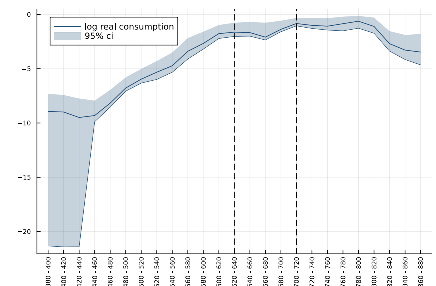
(f) Byzantine Heartlands



(g) al-Sham (Greater Syria)



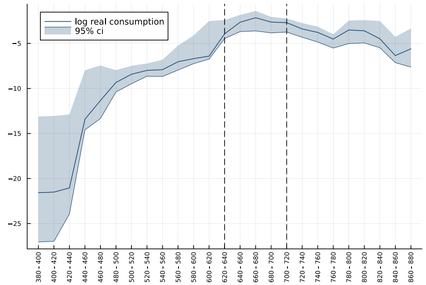
(h) Northern Syria and Caucasus



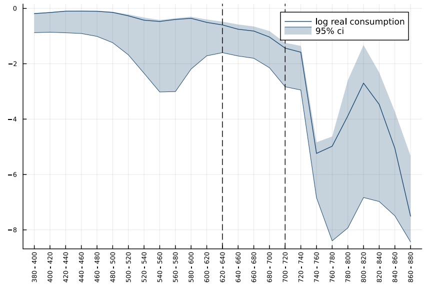
(i) al-Iraq, al-Jibal Khuzistan, Kirman

FIGURE B.6.—Real consumption AD 380-880

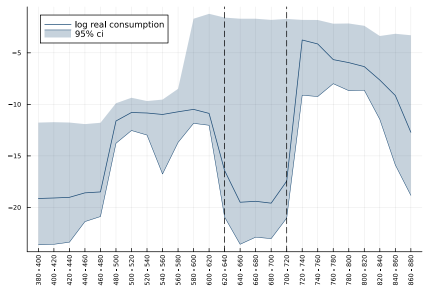
Notes: The figures on this page and the next complement figure 10, showing for all 13 regions the time-series from AD 380 to AD 880 of (log) real consumption as a share of aggregate consumption, using parameters from the specification in column 2 of table III. See section 5.1 for details. 95% confidence intervals (shaded areas) are computed from re-estimating our model on 100 bootstrapped samples from our coin hoard data. The vertical lines correspond to the Arab conquest, starting in 622 under Muhammad and ending in 713 with the conquest of Iberia.



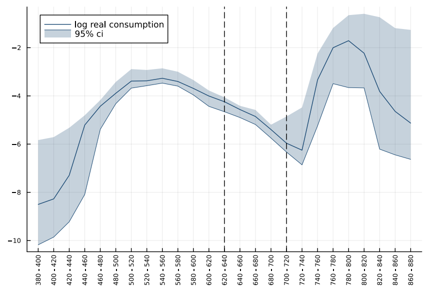
(j) Eastern Caliphate



(l) Misr (Egypt)



(k) Jazirat al-Arab and al-Yaman



(m) al-Maghrib

FIGURE B.6.—Real consumption AD 380-880 (continued)

C. ADDITIONAL THEORETICAL DERIVATIONS

We propose a simple extension of our model with both monetized transactions and barter.

Setup. We assume that a fraction $m_n[t]$ of transactions in region n and period t are monetized, and the remaining are barter: a fraction $m_n[t]$ of wages and of consumption are paid in coins. The dynamic market clearing condition (7) becomes,

$$(m_i[t]w_i\,[t]\,L_i\,[t]) = \sum_n \pi_{ni}\,[t]\,\Big((1-\lambda)\,(m_n[t-1]w_n\,[t-1]\,L_n\,[t-1]) + M_n\,[t]\Big), \forall(i,t). \tag{C.1}$$

The fraction of consumption $(X_n[t])$ spent in coins is,

$$m_n[t]X_n[t] = (1-\lambda)\,(m_n[t-1]w_n[t-1]L_n[t-1]) + M_n[t] \tag{C.2}$$

The dynamic of coin stocks remains unchanged, as in (9).

Following the intuition from Kiyotaki and Wright (1989), we assume that monetization increases with real production. We approximate this relationship with an iso-elastic function,

$$m_n[t] = m\left(T_n^{1/\theta}[t]L_n[t]\right) = \min\left\{\left(\frac{T_n^{1/\theta}[t]L_n[t]}{\overline{T^{1/\theta}L}}\right)^{\mu}, 1\right\}. \tag{C.3}$$

This parameterization captures several simple intuitions. First, there is a level of development $\overline{T^{1/\theta}L}$ above which an economy becomes fully monetized. We assume that none of the regions in our sample period have reached this stage of full monetization. Second, as in Kiyotaki and Wright (1989), a larger economy is characterized by more transactions, such that the benefit from using money is larger because it allows for more indirect (non-barter) exchanges.

We further assume that the income elasticity of monetization is equal to the dispersion of productivity between sectors in our Eaton and Kortum (2002) model of trade,

$$\mu = \theta. \tag{C.4}$$

This assumption captures in a reduced form the intuition in Kiyotaki and Wright (1989) that the degree of monetization depends on how beneficial the use of money is; with heterogeneous productivity between sectors, the benefits from using money depend on comparative advantages which govern the gains from multilateral exchanges, and therefore the gains from using a universally accepted medium of exchange. In the Eaton and Kortum (2002) ricardian model of trade, the parameter governing the dispersion of productivity across sectors is θ.

Bringing the model to the data. Because the dynamic of coin stocks remains unchanged compared to our baseline model (9), we follow the exact same maximum likelihood procedure as in section 4 to estimate the bilateral and seller terms, $d_{ni}^{-\theta}[t]$ and $\tilde{\beta}_i[t]$, and minting, $M_i[t]$. As in our baseline, we directly recover trade shares, $\pi_{ni}[t]$, from those parameter estimates.

The post-estimation computation of real consumption is different. We start with the dynamic market clearing condition with endogenous monetization (C.1). Knowledge of trade shares and minting is enough to recover *monetized* nominal incomes, $(m_n[t]w_n\,[t]\,L_n\,[t])$. Note that the solution for *monetized* incomes is identical to the solution for incomes in our baseline, although its interpretation is different.

We then perform a simple manipulation of the expressions for monetized consumption (C.2) and for real consumption similar to the manipulation in (23),

$$\frac{X_n[t]}{p_n[t]} = \gamma^{-1} \left(\pi_{nn}[t]\right)^{-1/\theta} \left(T_n^{1/\theta}[t] L_n[t]\right) \left(\frac{(1-\lambda)\left(m_n[t-1] w_n[t-1] L_n[t-1]\right) + M_n[t]}{\left(m_n[t] w_n[t] L_n[t]\right)}\right). \tag{C.5}$$

The first and last terms (openness and trade deficits) are directly computed from our solution for trade shares and monetized nominal incomes, and the trade elasticity θ.

To compute $T_n^{1/\theta}[t]L_n[t]$ we manipulate the seller term in the trade equation (11),

$$\begin{aligned}
\tilde{\beta}_n[t] &= \left(w_n[t]/T_n^{1/\theta}[t]\right)^{-\theta} \\
\Rightarrow T_n^{1/\theta}[t] L_n[t] &= w_n[t] L_n[t] \left(\tilde{\beta}_n[t]\right)^{1/\theta} \\
&= \left(m_n[t]\right)^{-1} \left(m_n[t] w_n[t] L_n[t]\right) \left(\tilde{\beta}_n[t]\right)^{1/\theta} \\
&= \left(\overline{T^{1/\theta} L}\right)^{\theta} \left(T_n^{1/\theta}[t] L_n[t]\right)^{-\theta} \left(m_n[t] w_n[t] L_n[t]\right) \left(\tilde{\beta}_n[t]\right)^{1/\theta} \\
\Rightarrow T_n^{1/\theta}[t] L_n[t] &= constant \times \left(\left(m_n[t] w_n[t] L_n[t]\right) \left(\tilde{\beta}_n[t]\right)^{1/\theta}\right)^{\frac{1}{1+\theta}}.
\end{aligned} \tag{C.6}$$

We can directly compute this last remaining term, productive capacity, using our solution for monetized nominal incomes, our estimates for seller terms, and the trade elasticity θ, up to a single (global) arbitrary multiplicative constant.

In practice this means that, compared to our baseline, we simply raise the productive capacity term labeled "$T_n^{1/\theta}[t]L_n[t]$" in the real consumption equation (23) to the power $1/(1+\theta)$ to account for endogenous monetization.

D. COIN HOARD DATA

Our numismatic data consists of two datasets: first, the set of hoards from the current release of the *Framing the Late Antique and Early Medieval Economy* project (FLAME, 2023). FLAME is a large collaborative effort of historians and numismatists that records data on coin hoards around the Mediterranean and Europe from between AD 325 and AD 725. We use the most recent release (January 2023) which has data on about 1.7m coins belonging to more than 9,000 hoards. Since the temporal and spatial focus of our study does not entirely overlap with that of FLAME, we complement their data by constructing a hand-coded dataset on hoards between AD 700 and AD 900, and hoards with a heavier emphasis on near eastern coins. We describe the hand-collected data and FLAME's data in turn.

D.1. *Hand-collected data*

We search the numismatic and archaeological literature for descriptions of coin hoards or coin finds with a *terminus post quem* (= date of the most recent content) of roughly between AD 700 and AD 950, that were discovered in Europe, North Africa, or the Middle East. For the sake of brevity we will refer to a single coin or a collection of coins that was found together in one place as a "hoard" (i.e. unless specifically mentioned, we do not distinguish between single finds, stray finds, mini-hoards, or full hoards). We exclude hoards that largely contain silver that was brought via the Viking route or that clearly have a Viking connection.[4] We likewise exclude records from excavations, unless they are described as a hoard or constitute a set of coins that were found together in the same location (e.g. in the same room of an excavated building).

An (at least approximate) findspot must be known for a hoard to be of use in our analysis. For each hoard we record the latitude and longitude of its findspot. When the findspot is known only with a low level of precision (e.g. at the country or region level) we code this in a separate dummy variable. Importantly, we do not record coins in museums or collections that have unknown findspots. While we digitize many descriptions of hoards that are incomplete, we omit hoards of which no information on the vast majority of coins has been published.

For each coin (or group of coins with identical properties) in a hoard we record, if documented by the authors of the hoard catalogue:

- The mint where the coin was minted, or believed to have been minted. When a coin is believed to have been an imitation, we note this separately.
- A time interval (consisting of a start year and end year) during which the coin was minted or is believed to have been minted. For some coins, such as most Islamic dirhams, this information is imprinted on the coin. For others, we code this as the shortest time interval during which the coin could have been minted, taking into account the denomination of the coin, the ruler under whose authority it has been issued, as well as his/her dynasty, and other information about coin types (e.g. pre/post-reform coinage). When the coin has been dated through the regnal year of the ruler or in the Islamic calendar, we convert this to Gregorian calendar years.

Beyond the attributes above, we record denomination, material, and issuing rulers and dynasties (mostly with dating of the coins in mind). This information, if known, is typically furnished by the authors of hoard catalogues in the numismatic literature. We do not distinguish between fragments and entire coins. Appendix D.5 gives the literature reference for each coin find that we hand-code. Some of these hoards fall outside our geographic and temporal focus, and will not be included in the estimation samples.

The geocodes of the hoards and mints are only approximate. We code Nomisma IDs for the mints based on the proximity of the place of minting, not based on the dynasty, e.g. “Siqilliyah” (Sicily) can be also used for non-Islamic issue.

D.1.1. *Byzantine hoards*

The hoards reported in the corpora by Pennas (1991), Füeg (2007), and Nikolaou and Touratsoglou (2019) form the basis of our collection of Byzantine hoards (the corpus on earlier finds by Morrisson, Popović, and Ivanišević (2006) is mostly already incorporated into FLAME). Information on particular regions come from Mirnik (1981) (Balkans), Arslan (2005) (Italy), Kovács (1989) (Hungary), and Wołoszyn (2009) (Central Europe). Hoard catalogues typically refer to collection catalogues (Sabatier, 1862, Wroth, 1908, Grierson, 1968, 1973) which we use to retrieve mint date intervals and likely mints.[5] We exclude coin finds from running excavations, unless the coins were found as individual parcels in a specific location. Table D.I shows our hand-coded byzantine hoards.

[4] Among the list from Appendix 3 of McCormick (2001), these are the hoards in Britain, Scandinavia, and Schleswig-Holstein (Germany). We also digitized the 10th century Máramaros county hoard (Fomin and Kovács, 1987), but drop it as its content (consisting of many imitations, as well as dirhams from the Samarkand and al-Shash mints) indicate that it was clearly brought in from the east.

[5] For a large part of the time interval that is not covered by FLAME, Byzantine gold and silver coins are believed to have been exclusively issued at Constantinople (Grierson, 1968).

D.1.2. *Hoards in the Near East and North Africa*

Table D.II shows the list of hand-coded hoards from the Near East and North Africa, along with references. These hoards consist mainly of Sasanian and/or Islamic coins, and sometimes Byzantine issue. We code approximate mint locations based on the proposals in the literature, typically giving preference to the suggestions of the authors of the original hoards.

A couple of notes on specific hoards:

- We digitize the Umm-Hajarah hoard based on the description by 'Ush (1972a) but follow Noonan (1980) in treating the isolated Seljuk coin that Al-'Ush dates to 689-690 AH as not belonging to the hoard.
- We digitize Hoge (1997)'s description of a hoard from "North Africa (or Spain?)", and assign Kairouan as approximate location (and note that the precise location of the hoard is immaterial to our exercise). We treat the Safavid dinar that is 650 years younger than the other coins (Hoge: "no doubt added to the other pieces 'in trade'.") as extraneous to the hoard.

D.1.3. *Islamic hoards in Spain and France*

Table D.III shows the hand-digitized hoards from Islamic Spain (al-Andalus) and Islamic coin finds from southern France.

D.1.4. *Other Islamic and Byzantine hoards in Europe*

We digitize the hoards, mini-hoards, and stray finds from McCormick (2001)'s survey of Arab and Byzantine coins in Europe (Appendix 3) between 668 and 900. We add those to our dataset, except when already covered in our other sources. We update hoard descriptions for which newer catalogues are available.[6] Finally, we exclude the contested Odoorn/Zuidbarge (1859–60) hoard, as the identity of it as a single hoard is not clear, some of the coins had been converted into jewellery, and the contents are not well described.[7]

D.1.5. *Carolingian hoards*

We follow Simon Coupland's *Checklist* (Coupland, 2011a, 2014, 2020) and digitize hoards and finds primarily based on the corpora presented by Völckers (1965), Duplessy (1985), and Haertle (1997), giving priority to more recent descriptions. Table D.IV shows details. We follow the mint codings of Louis the Pious' *Christiana religio* coins given by Coupland (2011b). As mentioned above, we exclude the contested Odoorn/Zuidbarge hoard.

D.2. *FLAME*

FLAME records their data in three different tables: coin finds, coin groups, and mints. In the coin find table each observation is a find that contains one or more coin groups; in the coin group table each observation is a set of coins with common recorded attributes (and linked to the coin find ID), including a mint and an interval for the year of minting. In the mint table each observation is a mint, and the mint name string allows these to be matched to coin groups. Mints and coin finds are geocoded.

[6] A35 (Steckborn): Ilisch (2005), A8 (Cagliari): Saccocci (2005), who also mentions an Aghlabid semi-dirham of Muhammad I found in Crotone, Sicily. We update A28 (Porto Torres, Sardinia) based on the number and datings reported in Füeg (2007)'s corpus, likewise the dates from A34 (Reno River).

[7] See Coupland (2011a) for a discussion of these issues.

The records in FLAME thus include a superset of the attributes in the hand-coded data above, except (i) the material of the coin, which we code based on the denomination; (ii) the weight and dimensions of the coins, which are sometimes (but not systematically) coded in the comments. We convert the FLAME data to the same structure as our handcoded data, including the following cleaning steps:

- A small number (6) of coin groups has a start year that's after the end year; we switch those around.
- FLAME contains start and end dates for the coin find itself. For a small number of coin groups the end date of the coin find falls in between the start and end dates of the coin group. This is often the case when very broad ranges have been given for the coin group, and so we truncate the coin group interval at the end date of the find.
- For Sasanian coins, we adhere to the mint codings in FLAME. A number of coins report the mint abbreviation but not the mint, we code and locate them analogously to how we coded them in the hand-coded coins (see below).
- A number of coin groups record a mint string that is not included in FLAME's mint file. We code Nomisma ID's for those mints, wherever possible.
- A large fraction of FLAME coins don't have mints or dates: often large hoards are not recorded by coin (just the total number of coins). Out of 1.7m coins, about 340k have mint and dates.

D.3. *Locating mints*

For FLAME data, we follow the attribution of mint locations done by the authors of the respective FLAME entries. For hand-collected data, we attempt to map the hand-coded mints to Nomisma (2023) IDs for the mints (`nmo:Mint`). Whenever a geocode for a mint is not available in Nomisma, or whenever the mint is not represented in Nomisma, we hand-code the geocodes. These geocodes should only be regarded as approximate and with a degree of precision required for our particular application in mind. Table D.V shows the mints we add to Nomisma, along with our codings, and Table D.VI shows the codings for existing Nomisma mints without geocodes.

D.3.1. *Sasanian mints*

The location of Sasanian mints and the identification of Sasanian mint signatures are contested. We generally follow the reading of the original hoard descriptions, except in situations where these are dated and the literature nowadays prefers different readings. Regarding the approximate location of the mints, we decided to code the approximate location for most signatures following the consensus in the literature; in some cases where the literature only agrees up to the region we chose Nomisma IDs from mints of that region. As with the other codings, the Nomisma IDs should only be seen as approximating the location of the mint, and do not carry any information on dating. Table D.VII summarizes our signature codings with their approximate mint locations.

D.4. *Political entities and the geography of hoards and mints*

D.4.1. *Dynasties/Empires*

We record dynasties/empires through the `dynastyname` field of FLAME data, and an equivalent field of the hand-coded data. We aggregate these to 10 more aggregate ("level 1") dynasties/empires, and seven most aggregate ("level 2") dynasties/empires. Figure D.1 shows the breakdown of recorded dynasties in our final sample.

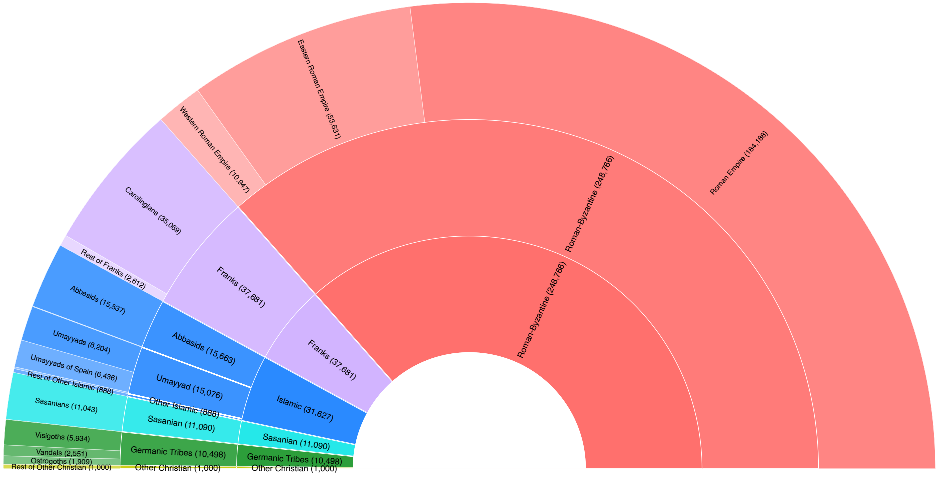


FIGURE D.1.—Dynasties/Empires

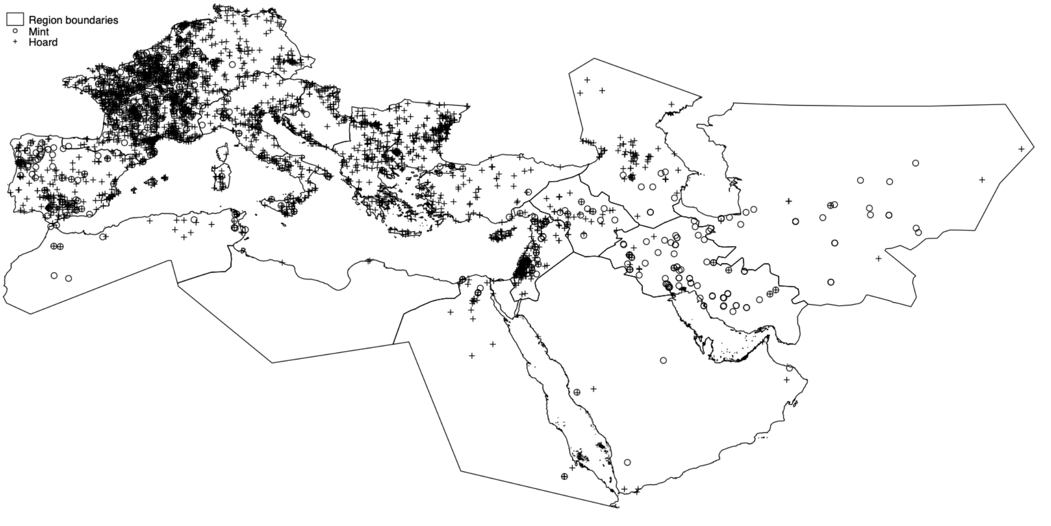
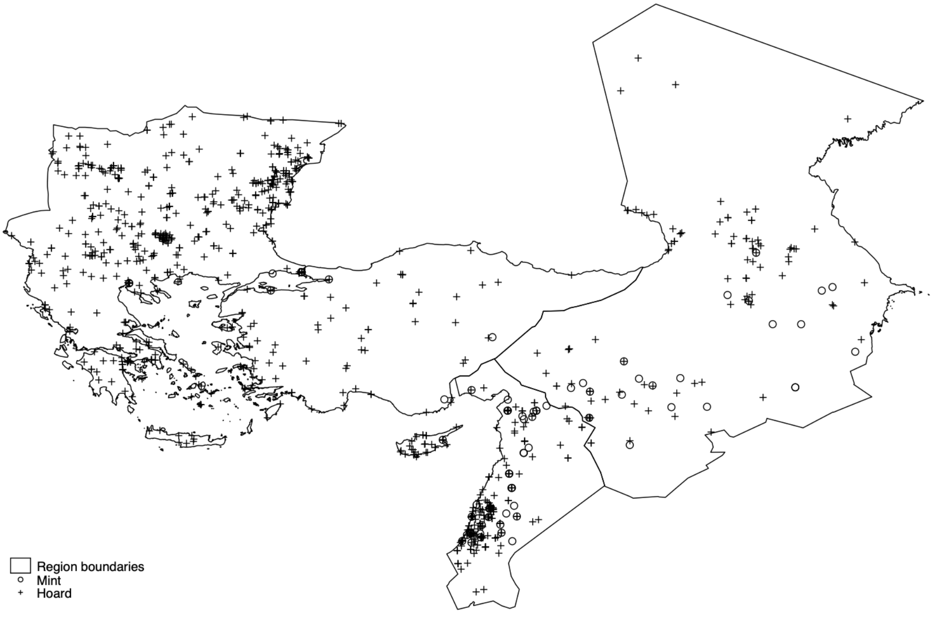


FIGURE D.2.—Mints and Hoards

D.4.2. *Location of hoards and mints*

Figure D.2 show the location of mints and hoards of our final dataset. Only locations corresponding to coins that were minted after AD 400 are shown. Figure D.3 shows details for western Europe and the eastern Mediterranean.

(a) Western Europe



(b) North-eastern Mediterranean

FIGURE D.3.—Mints and Hoards: Details

Note: Maps show coins minted after AD 400

D.5. *Tables and references*

TABLE D.I: Hoards with Byzantine coins

Hoard Name	Country	Reference	Lat.	Long.
Ankara 1960?	Turkey	Pennas (1991) 122	39.9388	32.8594
Argos 1983	Greece	Pennas (1991) 8	37.6353	22.7277
Ayies Paraskies/Crete 1962	Greece	Pennas (1991) 59 / Füeg (2007)	35.209	25.2041
Bajagic	Croatia	Mirnik (1981)	43.7581	16.6657
Balchik Stray Find I	Bulgaria	Curta (2005)	43.4119	28.1628
Berezeni	Romania	Oberländer-Târnoveanu (2001) p.67	46.378	28.1523
Bratimir	Bulgaria	Pennas (1991) 90	43.8682	26.7044
But	Italy	Arslan (2005) 2280	46.4768	13.0246
Byala 1954	Bulgaria	Pennas (1991) 73	42.8739	27.8886
Calarasi 1947	Romania	Pennas (1991) 111, Dimian (1957)	44.2029	27.3115
Camarina	Italy	Arslan (2005) 6170	36.8279	14.5241
Camarina ed. 18	Italy	Arslan (2005) 6185	36.8279	14.5241
Camarina ed. 1a	Italy	Arslan (2005) 6181	36.8279	14.5241
Camarina ed. 6	Italy	Arslan (2005) 6182	36.8279	14.5241
Capo Schiso 1950	Italy	Arslan (2005) 6910	37.8244	15.2684
Chryse/Edhessa 1935	Greece	Pennas (1991) 50 / Füeg (2007)	40.81	22.0446
Cleja	Romania	Pennas (1991) 113, Dimian (1957)	46.4019	26.9427
Constanta Stray I	Romania	Dimian (1957)	44.1777	28.6442
Constanta Stray II	Romania	Dimian (1957)	44.1777	28.6442
Corinth 15 May 1934 (South Basilica)	Greece	Pennas (1991) 3	37.9373	22.932
Corinth 1934	Greece	Pennas (1991) 7	37.9373	22.932
Corinth 1965 (Roman Bath)	Greece	Pennas (1991) 1	37.9373	22.932
Corinth 1965 (Roman Bath)	Greece	Pennas (1991) 4 (BCH 90, 1966, 751, 754)	37.9373	22.932
Corinth (St John's monastery)	Greece	Pennas (1991) 9	37.9373	22.932
Didyma (single find)	Turkey	Baldus (2006)	37.3731	27.2639
Drobeta - Turnu Severin	Romania	Oberländer-Târnoveanu (2001) p.68	44.6425	22.6587
Drobeta 1928	Romania	Oberländer-Târnoveanu (2001) p.67	44.6425	22.6587
Dubravice	Croatia	Mirnik (1981)	43.8506	15.9398
Dubrovnik 1982	Croatia	Mosser (1935) p.71 ("Ragusa"), Mirnik (1981) 359	42.6489	18.094
Elazig	Turkey	Füeg (2007)	38.6747	39.2229
Elbistan	Turkey	Füeg (2007)	38.2016	37.1924
Eskisehir	Turkey	Füeg (2007)	39.7743	30.5138
Gabrica	Bulgaria	Sophoulis (2011)	43.5082	26.9736
Govora	Romania	Oberländer-Târnoveanu (2001) p.68	45.0681	24.2302
Hadrianoupolis Acropolis Kimistene	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hadrianoupolis Basilica A	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hadrianoupolis Bath A	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hadrianoupolis Bath B	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hadrianoupolis Building 4	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hadrianoupolis Domus	Turkey	Laflı, Lightfoot, and Ritter (2016)	40.9231	32.4867
Hagios Nikolaos, Hydra (Greece)	Greece	Pennas (1996), p. 270	37.3011	23.3967
Iatrus 1962	Bulgaria	Pennas (1991) 77	43.6262	25.587
Iatrus 1975	Bulgaria	Pennas (1991) 75	43.6262	25.587
Ipsala	Turkey	Füeg (2007)	40.9201	26.3828
Istria Stray Finds 869-877	Croatia	Miškec (2002)	45.1439	13.8259
Kavakli	Turkey	Ünal (2018)	37.755	28.305
Kenchreai 1963	Greece	Pennas (1991) 2	37.8833	22.9873
Kozojedy, Bohemia	Czechia	Profantova (2009)	50.2548	13.8153
Kyme near Aliaga	Turkey	Carroccio, cited by Morrisson (2017)	38.7592	26.9367
Kyulevcha Grave	Bulgaria	Curta (2005)	43.2559	27.111
Lagbe	Turkey	Füeg (2007), Newell (1945)	36.8276	30.4112
Libice, Bohemia	Czechia	Profantova (2009)	50.1285	15.1815
Liopesi (around 1946)	Greece	Pennas (1991) 35 / Vryonis (1971)	37.9545	23.8521
Ljubimets	Bulgaria	Dimian (1957), Sophoulis (2011)	41.8466	26.0781
Luka Krnicka	Croatia	Miškec (2002)	44.9723	14.0171
Macvanska Mitrovica	Serbia	Pennas (1991) 72	44.9655	19.5975
Malthi (Dorion)	Greece	Pennas (1991) 6	37.267	21.8824
Maluk Povorets 1934	Romania	Pennas (1991) 74	43.7133	26.7652
Matera Piazza S. Francesco	Italy	Arslan (2005) 4140	40.6654	16.6087

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Hoard Name	Country	Reference	Lat.	Long.
Medias	Romania	Oberländer-Târnoveanu (2001) p.68, Dimian (1957)	46.1621	24.3567
Melito Porto Salvo	Italy	Arslan (2005) 0450	37.9197	15.7857
Mikulcice	Czechia	Profantova (2009)	48.8167	17.0516
Monemvasia Stray Find	Greece	Pennas (1996), p. 270	36.6876	23.0559
Naxos	Greece	Füeg (2007)	37.0567	25.4638
Nea Syllata/Chalkidiki 1977	Greece	Pennas (1991) 52	40.3275	23.136
Nin	Croatia	Mirnik (1981)	44.2392	15.1791
Odartsi	Bulgaria	Sophoulis (2011)	43.44	27.9616
Osava near Ram	Serbia	Füeg (2007)	44.8006	21.3433
Osvetimany	Czechia	Profantova (2009)	49.0562	17.2496
Oszony, Komarom	Hungary	Oberländer-Târnoveanu (2001) p.68	47.7295	18.1751
Piran	Italy	Arslan (2005) 2808	45.5279	13.5694
Pliska	Bulgaria	Füeg (2007)	43.362	27.1228
Prague, Tynsky dur	Czechia	Profantova (2009)	50.073	14.4286
Rakvice (Breclav)	Czechia	Profantova (2009)	48.8559	16.813
Rasova 1934	Romania	Pennas (1991) 112, Dimian (1957)	44.2403	27.9414
Reggio Calabria	Italy	Arslan (2005) 0670	38.0947	15.6455
Rhodos Stray Find 859	Greece	Kasdagli (2018)	36.436	28.2221
Rhodos V.12 (Kattavia)	Greece	Kasdagli (2018)	35.9534	27.7683
Rome / Tiber	Italy	Morrisson and Barrandon (1988)	41.8882	12.4768
Salamis (South of Amphitheatre, 1964-1974)	Turkey	Füeg (2007)	35.1914	33.8979
Santorini (Thira) 1895-1902	Greece	Pennas (1991) 57	36.4058	25.4588
Sicily (Fagerlie)	Italy	Fagerlie (1974)	37.5732	14.2114
Songurlu / Mosser	Turkey	Füeg (2007) / Mosser (1935)	40.1627	34.3767
Stare Mesto	Czechia	Profantova (2009)	49.0727	17.4463
Stimanga 1955	Greece	Pennas (1991) 5 (BCH 80, 1956, 256)	37.909	22.6989
Streda nad Bodrogom	Slovakia	Profantova (2009)	48.3785	21.758
Syracuse Via G. Di Natale	Italy	Arslan (2005) 7335	37.0724	15.2845
Tegani/Samos 1914	Greece	Pennas (1991) 58	37.6904	26.9417
Telerig Stray Miliaresion	Bulgaria	Curta (2005)	43.8457	27.671
Thessaloniki	Greece	Füeg (2007)	40.652	22.9304
Thessaloniki 1891	Greece	Pennas (1991) 51	40.652	22.9304
Tichilesti	Romania	Dimian (1957)	45.1291	27.9045
Tralleis/Aydin	Turkey	Ünal (2015)	37.8591	27.8335
Trilj	Croatia	Mirnik (1981)	43.6187	16.7241
Unknown Provenance (Turkey) 1987	Turkey	Pennas (1991) 123	39.2963	32.9327
Urluia 1936	Romania	Dimian (1957), Sophoulis (2011)	44.1016	27.9132
Velul lui Trajan	Romania	Pennas (1991) 105	44.1647	28.4621
Velul lui Trajan 1999/2000	Romania	Mănucu-Adameşteanu (2016)	44.1647	28.4621
Voila, Romania	Romania	Dimian (1957)	45.818	24.8405
Vukovar - Lijeva Bara	Croatia	Mirnik (1981)	45.3382	19.0079
Yakimovo (Progorelets) 1960	Bulgaria	Pennas (1991) 91	43.6337	23.3621
Yunak	Bulgaria	Pennas (1991) 76	43.0763	27.6109

TABLE D.II: Near East and North Africa Hoards

Hoard Name	Date	# Coins	Reference	Location	Lat.	Long.
Abu Saida	ca. 721	15	Society (1975)	Qaryat Abū Şaydā aş Şaghīrah, Iraq	33.924	44.761
Afaq	773-932	1674	Gachet (1993)	Afak, Iraq	32.064	45.247
Afghanistan	86-112 AH	131	Album (1971)	Afghanistan	33.000	66.000
Agrigenta	699-828	370	Lagumina (1904)	Agrigento, Italy	37.311	13.577
Al Raqqa	698-750	1187	Sears (2000)	Ar Raqqah, Syria	35.953	39.008
Al Wajh		35	Hakiem (1977)	Al Wajh, Saudi Arabia	26.246	36.452
Al-Khobar	tpq 784/85	42	Noonan (1980)	Khobar, Saudi Arabia	26.279	50.208
Amman	AH 79-125	12	Kirkbride (1951)	Amman, Jordan	31.955	35.945
Amūda I	tpq 874	646	Ilisch (1990)	ʻĀmūdā, Syria	37.104	40.930
Amūda II	779-941	643	Ilisch (1990)	ʻĀmūdā, Syria	37.104	40.930
Awarta (Nablus I)	602-685	29	Dajani (1951)	ʻAwartā, Palestine	32.161	35.284
Bab Tuma	tpq 748	854	Gyselen and Kalus (1983)	Damascus, Syria	33.510	36.291
Babylone, Egypt	157 AH - 241 AH	114	Jungfleisch (1949)	Cairo, Egypt	30.063	31.250
Buseyra	769-943	3108	Al Chomari (2020)	Al Buşayrah, Syria	35.156	40.427
Capernaum		288	Wilson (1989)	Kfar Nahum, Israel	32.881	35.575
Damascus	548-736	3815	’Ush (1972b)	Damascus, Syria	33.510	36.291
Damascus	679-721	546	’Ush (1954-1955)	Damascus, Syria	33.510	36.291
Denizbaji	tpq 811	2496	Artuk (1966)	Denizbacı, Turkey	37.139	38.390
Diyarbakir	802-902	224	Ilisch (1979)	Diyarbakır, Turkey	37.914	40.217
En Nebk	tpq 747	102	Royal Numismatic Society (1977)	An Nabk, Syria	34.024	36.728
Gazira	3rd to 9th century	2820	Gyselen and Nègre (1982)	Al Jazīrah, Iraq	36.000	42.000
Godhlaniya		127	American Numismatic Society (2023)	Syrian Arab Republic, Syria	35.000	38.000
Hamah	tpq 950	214	Ilisch (1990)	Hamāh, Syria	35.132	36.758
Iran 1970	tpq 820	668	Noonan (1980)	Islamic Republic of Iran, Iran	32.000	53.000
Isfahan	777-936	582	Lowick (1975)	Isfahan, Iran	32.652	51.675
Jarash		36	Treadwell and Rogan (1994)	Jarash, Jordan	32.281	35.899
Jazira (Illisch)	tpq 886	48	Ilisch (1990)	Al Jazīrah, Iraq	36.000	42.000
Kerman	about 632-651	43	Heidemann, Riederer, and Weber (2014)	Kerman, Iran	30.283	57.079
Khdir Elias	tpq 1014	2865	Al-Naqshbandi (1954)	Republic of Iraq, Iraq	33.000	44.000
Khorasan	705-774	196	Hebert (1966)	Mashhad, Iran	36.298	59.606
Khirbat al-Minya	716-734	2	Schneider (1952)	Horbat Minnim, Israel	32.865	35.536
Kufah	tpq 808/09	178	Noonan (1980)	Kufa, Iraq	32.051	44.440
Marv	tpq 815	855	Khodzhaniyazov and Treadwell (1998)	Mary, Turkmenistan	37.594	61.830
Near Fez		36	Royal Numismatic Society (1978)	Fès, Morocco	34.033	-5.000
Nippur (Bates)	704-794	76	Bates (1978)	Aţlāl Nafar, Iraq	32.136	45.221
Nippur (Sears)	597-743	97	Sears (1994)	Aţlāl Nafar, Iraq	32.136	45.221
North Africa (Spain?)	tpq 860	87	Hoge (1997)	Kairouan, Tunisia	35.678	10.096

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Hoard Name	Date	# Coins	Reference	Location	Lat.	Long.
Orif, Nablus	691-742	19	Ma'ayeh (1962)	Urif, Palestine, West Bank	32.159	35.224
Ouenza	789-798	12	Troussel (1942)	Ouenza, Algeria	35.953	8.129
Qamishliyyah	tpq 816	1519	Gyselen and Kalus (1983)	Al Qāmishlī, Syria	37.052	41.231
Ra's al-Khaimah	921-975	43	Lowick and Nisbet (1968)	Ras Al Khaimah City, UAE	25.790	55.943
Sinaw	589-841	948	Lowick (1983)	Sināw, Oman	22.501	58.030
Tabaristan	about 718-760	810	Malek (1996)	Mazandaran Province, Iran	36.250	52.333
Tiflis	ca. 280-330 AH	112	Bartolomei (1857)	Tbilisi, Georgia	41.694	44.834
Umm Hajarah	tpq 808/09	408	'Ush (1972a)	Umm Hajarah, Syria	36.195	41.074
Utaifiyah	154-193 AH	294	Bakri (1973)	Baghdad, Iraq	33.341	44.401
Volubilis	tpq 125 AH (742)	232	Eustache (1956)	Oualili, Morocco	34.073	-5.555
Yarubiyyah	tpq 815/816	1415	American Numismatic Society (2023)	Al Ya'rubīyah, Syria	36.811	42.062
Zahu/Zakho	tpq 808-9	3306	Al-Naqshbandi (1949, 1950, 1951, 1952)	Zaxo, Iraq	37.149	42.686

TABLE D.III: Hoards in al-Andalus

Hoard name	Date	# Coins	Reference	Location	Latitude	Longitude
Alcaudete	698-734	14	Cano Ávila (1989)	Alcaudete	N 37° 35' 27"	W 4° 4' 56"
Algeciras	710-727	29	Canto García and Martín Escudero (2009)	Algeciras	N 36° 7' 59"	W 5° 27' 1"
Alhama	770-876	459	Zaidín (1892)	Alhama	N 37° 0' 24"	W 3° 59' 22"
Arrabal Occidental	929-1021	373	Canto García, Clapés, and Jabłońska (2020a)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Azanuy	699-733	6	Zaidín (1913)	Azanuy	N 41° 59' 10"	E 0° 18' 58'
Badajoz	927-1011	99	Prieto (1934)	Badajoz	N 38° 52' 40"	W 6° 58' 14"
Baena	699-754	160	Martín Escudero (2001)	Baena	N 37° 39' 22"	W 4° 20' 4"
Barrio de los Olivos Borrachos	941-1004	165	Marcos Pous and Vicent Zaragoza (1992)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Benferri	941-958	12	Doménech Belda (1997)	Benferri	N 38° 8' 28"	W 0° 57' 43"
Bormujos	929-965	11	Cano Ávila (2016)	Bormujos	N 37°21'41.9"	W 6° 06' 38.1"
Calle San Jose	936-950	16	Doménech Belda (1997)	Xàtiva	N 38° 59' 25"	W 0° 31' 6"
Calle San Pedro	967-1031	19	Canto García and Jablońska (2019)	Murcia	N 37° 59' 13"	W 1° 7' 48"
Calle Santa Julia	929-1012	263	Segovia Sopo (2014)	Mérida	N 38° 54' 58"	W 6° 20' 37"
Campo de la Verdad	775-912	176	Martín and Martín (2006)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Carmona	698-753	146	Canto García and Escudero (2012)	Carmona	N 37°28' 17"	W 5°38' 46"
Castillejos de Quintana	933-1010	39	Cravioto (2016)	Castillejos de Quintana	N 36°46'58.7"	W 4° 41' 30.9"
Castro Marim	788-885	53	Rodrigues Marinho (1995)	Castro-Marim	N 37° 13' 14"	W 7° 26' 36"
Cerro da Villa	831-900	239	Heidemann, Schierl, and Teichner (2018)	Cerro da Villa	N 37° 4' 48"	W 8° 7' 13"
Crevilllent	770-1269	34	Doménech Belda and Trelis (1990)	Crevillent	N 38° 14' 59"	W 0° 48' 35"
Cihuela	912-1016	296	Palacios (1961a)	Cihuela	N 41° 24' 26"	W 1° 59' 59"
Consuegra	835-1010	173	Martín Escudero (2011)	Consuegra	N 39° 27' 44"	W 3° 36' 28"
Cordoba I	817-1010	25	Palacios (1961b)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Cordoba II	933-953	328	Palacios (1958)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Cordoba III	933-1021	379	Palacios (1958)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Cordoba IV	708-796	119	Canto García (1988)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Cova del Randerro	768-835	54	Doménech Belda (1997)	Pedreguer	N 38° 47' 35"	E 0° 2' 2"
Cuba	932-1010	9	Martín Escudero (2011)	Cuba	N 38° 10' 24"	W 7° 53' 46"
Domingo Perez	767-865	367	Martín and Martín (2002)	Domingo Pérez	N 37° 29' 45"	W 3° 30' 33"
Elche	841-1173	316	Doménech Belda (1992)	Elche	N 38° 15' 43"	W 0° 42' 3"
Electromecanicas I	941-1005	169	Marcos Pous and Vicent Zaragoza (1992)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
Electromecanicas II	928-1016	102	Marcos Pous and Vicent Zaragoza (1992)	Cordoba	N 37° 53' 29"	W 4° 46' 21"
El Pedroso	928-1021	144	Cano Ávila and Martín Gómez (2006)	Hacienda Montegil, El Pedroso	N 37°43'51.9"	W 5°51'39.8"
El Pedroso III	832-1021	144	Cano Ávila and Gómez (2008)	El Pedroso	N 37° 51' 0"	W 5° 46' 0"

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Hoard name	Date	# Coins	Reference	Location	Latitude	Longitude
El Rebollar	810-818	5	SDGLGOGA+ (2020)	Boalo	N 40° 42’ 57”	W 3° 54’ 59”
Finca la Marquesa	941-1036	246	Doménech Belda (1997)	Montilla	N 37°36’07.2"	W 4°37’11.7"
Fontanar	941-977	764	Canto García and Martín Escudero (2007)	Cordoba	N 37° 53’ 29”	W 4° 46’ 21”
Fuente de Cantos	837-883	15	Segovia Sopo (2006)	Fuente de Cantos	N 38° 15’ 0”	W 6° 18’ 0”
Hospital Militar	970-1032	23	Martín Escudero (2003)	Zaragoza	N 41° 39’ 21”	W 0° 52’ 38”
Huesca	710-756	100	Martín Escudero (2012)	Huesca region		
Izcar	778-886	50	Ariza Armada (1988)	Cortijo de Izcar	N 37°39’56.1"	W 4°23’41.6"
Iznajar	768-912	1047	Canto García and Marsal Moyano (1988)	Iznajar	N 37° 15’ 27”	W 4° 18’ 30”
Jaen	711-713	4	González García and Martínez Chico (2017)	Jaen region		
Jerez de los Caballeros	770-782	277	Canto García (2019)	Jerez de los Caballeros	N 38° 19’ 14”	W 6° 46’ 21”
La Almagra	820-822	7	Museo Arqueológico de Murcia (2014)	La Almagra	N 38° 2’ 15”	W 1° 25’ 57”
La Fuensanta	770-812	18	Cravioto and Ayala (1995)	Cerro la Fuensanta	36°55’13.7"N	4°23’23.7"W
Lantejuela	773-887	175	Ruiz Asencio (1967)	La Lantejuela	N 37°19’17.5"	W 5°13’27.6"
La Rinconada	770-912	315	Cano Ávila and Martín Gómez (2005)	La Rinconada	N 37° 29’ 10”	W 5° 58’ 51”
Las Torres	757-976	18	Martínez Enamorado (2004)	Gavilanes	N 40° 15’ 44”	W 4° 51’ 30”
L’Elca	933-950	31	Doménech Belda (1997)	Oliva	N 38° 55’ 10”	W 0° 7’ 9”
Lleida	770-1463	40	Soler Balaguero (1993)	Lleida	N 41° 37’ 7”	E 0° 34’ 29”
Lora del Rio	941-1021	165	Bru (1985)	Lora del Rio	N 37° 39’ 32”	W 5° 31’ 39”
Los Villares	942-1028	112	Valle (1987)	Caudete de las Fuentes	N 39° 33’ 34”	W 1° 16’ 42”
Madinat Iyyuh	711-856	20	Doménech Belda and Gutiérrez Lloret (2006)	El Tolmo de Minateda	N 38° 28’ 34”	W 1° 36’ 20”
Marroquies Altos	933-1010	270	Asencio (1962)	Jaen	N 37° 46’ 9”	W 3° 47’ 25”
Marroquies Bajos	941-1015	201	Canto García, García Ruiz, and Ruiz Quintanar (1997)	Jaen	N 37° 46’ 9”	W 3° 47’ 25”
Martos	817-875	24	Canto García (1993)	Cortijo del Mimbre	N 37°38’26.0"	W 3°57’04.9"
Merida	726-901	60	Rodríguez Palomo and Martín Escudero (2022)	Merida	N 38° 54’ 58”	W 6° 20’ 37”
Mertola	932-1036	81	Poiares (2000)	Mertola	N 37° 38’ 34”	W 7° 39’ 40”
Mijas Costa	932-976	533	Ayala Ruiz and Gozalbes Cravioto (1996)	La Cala de Mijas	N 36° 33’ 56”	W 4° 40’ 11”
Montellano	949-1010	23	Cano Ávila (2014)	Montellano	N 37°00’06.1"	W 5°33’02.1"
Moraleja	767-854	16	Álvarez (1993)	Moraleja	N 40° 0’ 58”	W 6° 41’ 51”
Moreria	857-1015	134	Palma García and Segovia Sopo (2007)	Merida	N 38° 54’ 58”	W 6° 20’ 37”
Niebla	805-884	36	Cano Ávila and Martín Gomez (2011)	Sierra de Alcantara	N 37°28’33.8"	W 6°38’36.2"
Osuna	954-1022	3	Alfaro Asins (1992)	Osuna	N 37° 14’ 15”	W 5° 6’ 11”
Parque Cruz conde	852-1021	3341	Canto García, Martín Escudero, and Jablonska (2020b)	Cordoba	N 37° 53’ 29”	W 4° 46’ 21”
Partida de Atzbares	941-970	26	Doménech Belda (1997)	Atzavares Baix (Elche)	N 38° 15’ 43”	W 0° 42’ 3”
Pascul de Gayangos	778-1204	159	Marinho (1993)	Algarve		
Pinos Puente	770-816	169	Martín Escudero (2011)	Pinos Puente	N 37° 15’ 3”	W 3° 44’ 58”

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Hoard name	Date	# Coins	Reference	Location	Latitude	Longitude
Pozoblanco	948-976	15	Marcos Pous and Vicent Zaragoza (1992)	Pozoblanco	N 38° 22’ 44”	W 4° 50’ 53”
Priego de Cordoba	770-856	54	Ávila and Pareja (1999)	Priego de Cordoba	N 37° 26’ 17”	W 4° 11’ 42”
Puebla de Cazalla	770-892	911	Ibrahim and Canto García (1991)	La Puebla de Cazalla	N 37° 13’ 17”	W 5° 18’ 41”
Puente de Miluze	934-1057	164	Canto García (2001)	Pamplona	N 42° 49’ 0”	W 1° 38’ 35”
Recopolis	772-785	9	Priego and Enciso (2016)	Recopolis	N 40°19’15.1"	W 2°53’37.7"
Sagrada Familia	945-1012	316	Marcos Pous and Vicent Zaragoza (1992)	Cordoba	N 37° 53’ 29”	W 4° 46’ 21”
San Andres de Ordoiz	782-908	167	Uranga (1950)	Estella-Lizarra	N 42°40’ 19”	W 2°01’ 56”
Saqunda	707-930	467	Martín Escudero, Casal García, and Canto García (2023)	Cordoba	N 37° 53’ 29”	W 4° 46’ 21”
Sevilla	711-1011	497	Saenz-Diéz (1993)	Sevilla	N 37° 22’ 58”	W 5° 58’ 23”
Sierra Cazorla	928-1021	237	Bru (1982)	Sierra Cazorla	N 37° 54’ 45”	W 2° 58’ 34”
Silves	770-875	79	Miles (1960)	Silves	N 37° 11’ 21”	W 8° 26’ 17”
Sinarcas	942-1037	57	Arroyo Ilera (1989)	Sinarcas	N 39° 44’ 0”	W 1° 14’ 0”
Solar del Museo Arqueologico	953-1007	16	Marcos Pous and Vicent Zaragoza (1992)	Cordoba	N 37° 53’ 29”	W 4° 46’ 21”
South France	692-886	204	Parvérie (2014, 2019)	South France		
Spain single finds (felus)	699-901	57	Martín Escudero (2012)	Spain		
Tarancon	929-1014	451	Canto García (2014)	Tarancon	N 40° 0’ 30”	W 3° 0’ 26”
Teatro romano	805-819	25	Segovia Sopo and Jiménez (2011)	Merida	N 38° 54’ 58”	W 6° 20’ 37”
Tignar	864-913	35	Motos Guirao and Díaz García (1985)	Albolote	N 37° 13’ 51”	W 3° 39’ 18”
Tijan	976-1021	377	Fontenla Ballesta (1998)	Sierra de Cabrera	N 37°06’30.3"	W 1°55’49.2"
Trujillo	711-1014	384	Palacios (1957)	Trujillo	N 39° 27’ 28”	W 5° 52’ 55”
Valencia de Ventoso	933-1006	7	Grañeda Miñón (2021)	Valencia del Ventoso	N 38° 16’ 0”	W 6° 28’ 0”
Valeria	936-1009	250	Puertas (1982)	Valeria	N 39° 47’ 0”	W 2° 9’ 0”
Valle de Guadajoz	931-1013	204	Ortega, Estella, Torres, and Marqués (2006)	Fuentiduena (Baena)	N 37°43’42.1"	W 4°16’54.3"
Vega Baja	-200-1500	184	Priego (2020)	Toledo	N 39° 51’ 29”	W 4° 1’ 21”
Vera	941-1024	370	Doménech Belda (1997)	Vera	N 37° 14’ 36”	W 1° 51’ 32”
Villaviciosa	705-817	1361	Peña Martín and Vega Martín (2007)	Villaviciosa de Cordoba	N 38° 5’ 0”	W 5° 1’ 0”
Yecla	705-726	5	Zaidín (1913)	Yecla	N 38° 39’ 18”	W 1° 7’ 46”
Zafra	789-892	43	Canto García (2019)	Zafra	N 38° 25’ 31”	W 6° 25’ 2”
Zamora	943-999	10	Cerrato and Esquivel (2019)	Zamora	N 41° 30’ 22”	W 5° 44’ 40”

TABLE D.IV: Carolingian Hoards

Hoard Name	Date	Reference	Location	Latitude	Longitude
Aalst	840-855	Bijsterveld, Neufeglise, and Veen (2000)	Aalst	51.39611	5.477
Aalsum	814-855	Morrison and Grunthal (1967)	Aalsum	53.3403	6.00538
Achlum	768-840	Morrison and Grunthal (1967)	Achlum	53.14779	5.48239
Alfocea	943-977	Parvérie (2018)	Alfocea	41.724097	-0.953131
Amerongen	768-877	Coupland (2014)	Amerongen	52.0025	5.46024
Ampurias	768-814	Doménech-Belda and others (2013)	Ampurias	42.134477	3.111418
Andalusia	814-848	Parvérie (2018)	Andalusia		
Angeac-Champagne	840-877	Duplessy (1985)	Angeac-Champagne	45.60769	-0.29771
Angers I	814-840	Morrison and Grunthal (1967)	Angers	47.4707	-0.55324
Angers II (Saint-Julien)	819-877	Haertle (1997)	Angers	47.4707	-0.55324
Anglure	864-887	Morrison and Grunthal (1967)	Anglure	48.58345	3.81356
ANS find	768-922	Morrison and Grunthal (1967)	France		
Anse I	818-823	Guillemain (1993)	Anse	45.937639	4.717512
Anserall	768-815	Doménech-Belda and others (2013)	Anserall	42.37829	1.456511
Apremont	793-822	Morrison and Grunthal (1967)	Apremont-sur-Allier	46.906	3.048
Aquitaine	814-887	Coupland (1991)	Aquitaine		
Ardres	888-923	Haertle (1997)	Ardres	50.856432	1.978355
Arras	843-922	Morrison and Grunthal (1967)	Arras	50.29039	2.778414
Ashdon	843-898	Blackburn (1989)	Ashdon	52.05544	0.31373
Aspres-lès-Corps	901-924	Schulze (1984)	Aspres-lès-Corps	44.80162	5.98217
Assebroek	843-877	Morrison and Grunthal (1967)	Assebroek	51.18793	3.27363
Assen	800-911	Morrison and Grunthal (1967)	Assen	52.99421	6.55957
Auxerre	813-877	Haertle (1997)	Auxerre	47.796587	3.570535
Auzeville	814-848	Sarah, Geneviève, and Guerrot (2016)	Auzeville	43.5257	1.49342
Avallon	843-877	Coupland (2020)	Avallon	47.488712	3.907758
Avignon	843-887	Morrison and Grunthal (1967)	Avignon	43.95344	4.80601
Bakonyszombathely	898-973	Morrison and Grunthal (1967)	Bakonyszombathely	47.47208	17.96018
Balloo	843-855	Haertle (1997)	Balloo	54.472363	-5.69076
Barbentane	814-840	Morrison and Grunthal (1967)	Barbentane	43.89948	4.74635
Barcelona	814-840	Doménech-Belda and others (2013)	Barcelona	41.395937	2.174552
Bassenheim	814-876	Coupland (2019)	bassenheim	50.359028	7.462443
Bátorove Kosihy	888-950	Kovács (1989)	Bátorove Kosihy	47.83083	18.41083
Beaumont	843-877	Morrison and Grunthal (1967)	Beaumont (Chalo Saint Mars)	48.409016	2.042742
Bel-Air	768-814	Morrison and Grunthal (1967)	Lausanne	46.57957	6.605807
Bellpuig	887-928	Doménech-Belda and others (2013)	Bellpuig	41.626531	1.011607
Belvézet	768-840	Morrison and Grunthal (1967)	Belvézet	44.08433	4.36426
Bikbergen	814-855	Cruysheer and Veen (2015)	Bikbergen	52.287933	5.196186
Bjerndrup	817-924	Coupland (2020)	Bjerndrup	54.93391	9.32867
Blendecques	814-840	Coupland (2020)	Blendecques	50.716982	2.282169
Bligny	814-887	Morrison and Grunthal (1967)	Bligny	48.1725	4.6172
Blois	898-940	Moesgaard (1997)	Blois	47.58696	1.33139
Bondeno	768-814	Morrison and Grunthal (1967)	Bondeno	44.89098	11.41096
Bonnevaux	800-887	Morrison and Grunthal (1967)	Bonnevaux	44.367837	4.030289
Borne	794-813	Coupland (2011a)	Borne	52.30137	6.75779
Bourges	840-877	Morrison and Grunthal (1967)	Bourges	47.08585	2.39293
Bourges	800-887	Coupland (2020)	Bourges	47.08585	2.39293
Bourgneuf	814-888	Morrison and Grunthal (1967)	Bourgneuf	46.167624	-1.022216
Bourgneuf-en-Retz	843-877	Coupland (2010)	Bourgneuf-en-Retz	47.04229	-1.9543
Bray-sur-Seine	840-877	Vandenbossche and Coupland (2012)	Bray-sur-Seine	48.41451	3.24057
Bressuire	814-840	Coupland (1995)	Bressuire	46.84008	-0.49253
Breuvery-sur-Coole	768-813	Dhénin (1989)	Breuvery-sur-Coole	48.86311	4.31164
Brion	814-840	Denais (1908)	Brion	47.4425	-0.1553
Brioux-sur-Boutonne	814-840	Morrison and Grunthal (1967)	Brioux-sur-Boutonne	46.14349	-0.21823
Bruère-Allichamps	814-954	Morrison and Grunthal (1967)	Bruère-Allichamps	46.7695	2.4325
Burgum	843-877	Haertle (1997)	Burgum	53.19527	5.98694
Caden	843-877	Coupland (2020)	Caden	47.630822	-2.287131
Caen	936-954	Coupland (2020)	Caen	49.183512	-0.363489
Calatrava la vieja		Parvérie (2018)	Calatrava la Vieja	39.074099	-3.833274
Campeaux	813-877	Haertle (1997)	Campeaux	48.952844	-0.93197
Carcassonne	768-814	Coupland (2014)	Carcassonne	43.206463	2.363268
Castelsarasin	888-898	Morrison and Grunthal (1967) and Lafaurie (1965)	Castelsarasin	44.039071	1.106969

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Hoard Name	Date	Reference	Location	Latitude	Longitude
Catalonia	768-905	Balaguer (1999) and Doménech-Belda and others (2013)	Calalonia		
Cauroir	843-882	Coupland (2011a)	Cauroir	50.17283	3.30174
Cerdanyola	814-840	Doménech-Belda and others (2013)	Cerdanyola	41.49201	2.137338
Cerveník	826-950		Cerveník	48.45	17.75
Chaley	936-954	Morrison and Grunthal (1967)	Chaley	45.9552	5.53122
Chalo-Saint-Mars	840-877	Morrison and Grunthal (1967)	Chalo-Saint-Mars	48.4267	2.067
Chalon-sur–Saône I	800-887	Morrison and Grunthal (1967)	Chalon-sur-Saône	46.782132	4.858459
Chalon-sur-Saone II	800-887	Haertle (1997)	Chalon-sur-Saône	46.782132	4.858459
Charente-Maritime	888-898	Coupland (2011a)	Charente-Maritime		
Chartes	923-977	Duplessy (1985)	Chartres	48.446659	1.488596
Chartres II	751-768	Morrison and Grunthal (1967)	Chartres	48.446659	1.488596
Château Roussillon	793-877	Haertle (1997)	Château Roussillon	42.710278	2.946667
Chateauneuf sur Cher	843-954	Morrison and Grunthal (1967)	Chateauneuf sur Cher	46.857333	2.320522
Chaumoux-Marcilly	814-877	Morrison and Grunthal (1967)	Chaumoux-Marcilly	47.12628	2.77884
Chauvigny	843-877	Société des antiquaires de l'Ouest (1982)	Chauvigny	46.56974	0.64345
Chef-Boutonne	800-922	Haertle (1997) and Rondier (1869)	Chef-Boutonne	46.10934	-0.06806
Chester	888-924	Webster, Dolley, and Dunning (1953)	Chester	53.1903	-2.89437
Chézy-sur-Marne	768-814	Duplessy (1985)	Chézy-sur-Marne	48.989611	3.366294
Choisy-au-Bac	888-898	Haertle (1997)	Choisy-au-Bac	49.44777	2.88097
Ciney Dinant	898-922	Coupland (2020)	Ciney	50.286773	5.098966
Clermont Ferrand	843-918	Coupland (2020)	Clermont-Ferrand	45.778063	3.083696
Compiègne I	877-882		Compiègne	49.41762	2.82513
Compiègne II	843-882	Morrison and Grunthal (1967)	Compiègne	49.41762	2.82513
Corrèze	843-877	Coupland (2014)	Corrèze		
Cosne d'Allier	814-840	Coupland (2014)	Cosne d'Allier	46.474799	2.830127
Cosne-Cours-sur-Loire II	814-877	Morrison and Grunthal (1967)	Cosne-Cours-sur-Loire	47.40983	2.92425
Cosne-Cours-sur-Loire III	877-840	Haertle (1997)	Cosne-Cours-sur-Loire	47.40983	2.92425
Croydon	814-877	Morrison and Grunthal (1967)	Croydon	51.379287	-0.09975
Csorna	888-947	Kovács (1989)	Csorna	47.6167	17.25
Cuerdale	843-922	Morrison and Grunthal (1967)	Cuerdale	53.7553	-2.638
Dalen	843-976	Morrison and Grunthal (1967)	Dalen	52.69847	6.75641
Dauphiné	814-848	Coupland (2014)	Dauphiné		
Deux-Sèvres	814-877	Société de statistique, sciences, lettres et arts du département des Deux-Sèvres (1882)	Deux-Sèvres		
Dijon	770-780	Bompaire and Depierre (1989)	Dijon	47.3268	5.04619
Dommartin-Lettrée	923-936	Duplessy (1985)	Dommartin-Lettrée	48.7669	4.29933
Dordives	750-950	Coupland (2014)	Dordives	48.144081	2.766333
Dorestad	768-877	Morrison and Grunthal (1967)	Dorestadt	51.97212	5.344769
Drantum	814-840	Haertle (1997)	Drantum	52.81942	8.19537
Eichstetten	911-922	Morrison and Grunthal (1967)	Eichstetten	48.094296	7.745429
Ejstrup	814-840	Coupland (2020)	Ejstrup	55.503525	9.377413
Ekeren	819-877	Haertle (1997)	Ekeren	51.276405	4.417467
Ellikon an der Thur	887-915	Zäch (2001)	Ellikon an der Thur	47.56253	8.82386
Emmen	814-877	Morrison and Grunthal (1967)	Emmen	52.49784	6.23039
Entrammes	814-877	Coupland (2014)	Entrammes	47.999133	-0.716154
Espana 1-4	800-1009	Parvérie (2018)	Calatayud	41.352868	-1.641101
Etampes	843-882	Morrison and Grunthal (1967)	Etampes	48.434768	2.162027
Etréchy	832-877	Morrison and Grunthal (1967)	Etrechy	48.88411	3.94374
Evreux	840-954	Duplessy (1985) and Moesgaard (2003)	Evreux	49.02754	1.15028
Extremadura		Parvérie (2018)	Extremadura		
Eyguières	814-840	Coupland (2020)	Eyguières	43.696133	5.030134
Fécamp	900-999	Duplessy (1985)	Fécamp	49.75765	0.37632
Flacey	814-840	Coupland (2020)	Flacey	48.147247	1.349598
Flanders	814-877	Coupland (2020)	Flanders		
Florange		Duplessy (1985) and Simmer (2000)	Florange	49.32743	6.12273
Foissy-lès-Vézelay	864-877		Foissy-lès-Vézelay	47.43637	3.76447
Fontaines	814-877	Duplessy (1985)	Fontaines	46.85083	4.773055
Frankfurt	814-840	Morrison and Grunthal (1967)	Frankfurt am Main	50.11208	8.68341
Freiburg im Breisgau	898-922	Morrison and Grunthal (1967)	Freiburg im Breisgau	47.99853	7.84965
Fresnes		Duplessy (1985)	Fresnes	48.75043	2.322063
Fridolfing	768-814	Coupland (2020)	Fridolfing	47.998573	12.826917
Frisia	814-855	Morrison and Grunthal (1967)	Grou	53.11035	5.848604
Gannat	800-887	Morrison and Grunthal (1967)		46.10192	3.19692
Gelderland	768-814	Morrison and Grunthal (1967)	Gelderland		
Giekau	814-911	Wiechmann (2004)	Giekau	54.31793	10.50529

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Hoard Name	Date	Reference	Location	Latitude	Longitude
Glisy	800-922	Morrison and Grunthal (1967)	Glisy	49.8756	2.39788
Gnadendorf	898-905	Daim and Lauermann (2006)	Gnadendorf	48.61549	16.39885
Goutum	814-877	Coupland (2020)	Goutum	53.178037	5.806018
Grisebjerggård	898-922		Slagelse	55.3028	11.2647
Groningen	814-877	Morrison and Grunthal (1967)	Groningen	53.25713	6.93525
Guardamiglio	843-884	Coupland (2011a)	Guardamiglio	45.11055	9.68215
Györ I	888-950	Kovács (1989)	Györ	47.69739	17.6527
Györ II	888-951	Kovács (1989)	Györ	47.69739	17.6527
Halimba	902-947	Kovács (1989)	Halimba	47.03345	17.53546
Häljarp	814-840	Morrison and Grunthal (1967)		55.85578	12.910919
Harkirke	843-905	Morrison and Grunthal (1967)	Crosby	53.48919	-3.048081
Harlingen	840-855	Haertle (1997)	Harlingen	53.1735	5.4246
Haute Isle	814-922	Morrison and Grunthal (1967)	Haute Isle	49.083426	1.65697
Haza de Carmen	888-954	Coupland (2020)	Cordoba	37.881495	-4.776125
Hermenches	822-840	Morrison and Grunthal (1967)	Hermenches	46.640456	6.757567
Hoen	814-855	Morrison and Grunthal (1967)	Hoen	60.2204	10.25852
Hole	796-840	Coupland (2020)	Hole	58.897156	6.018229
Holy Family	800-887	Parvérie (2018) and Morrison and Grunthal (1967)	Cordoba	37.888028	-4.7734
Hradec Hilfort	768-814	Coupland (2020)	Hradec-Kralove	50.209703	15.832231
Huriel	800-887	Morrison and Grunthal (1967)	Le Moulin-Gargot (Huriel)	46.37468	2.47842
Ibaneta	800-888	Doménech-Belda and others (2013)	Puerto d'Ibaneta	43.020083	-1.324207
Ibersheim	768-814	Morrison and Grunthal (1967)	Ibersheim	49.72085	8.40065
Ilanz I	843-905	Morrison and Grunthal (1967)	Ilanz	46.77451	9.20463
Ilanz II	664-814	Bernareggi (1977, 1983), Völckers (1965), McCormick (2001)	Ilanz	46.77451	9.20463
Île Agois	864-877	Johnston (1986)	Île Agois	49.24935	-2.18641
Île-de-France	888-936	Dhénin (2006)	Ile de France		
Imbleville	864-877	Haertle (1997)	Imbleville	49.71539	0.95198
Imphy	751-814	Morrison and Grunthal (1967)	Imphy	46.934537	3.259903
Indre	814-865	Morrison and Grunthal (1967)	Indre		
Indre II	814-848	Coupland (2014)	Indre		
Indre-et-Loire	814-877	Coupland (2011a)	Indre-et-Loire		
Indre-et-Loire II	888-910	Coupland (2011a)	Indre-et-Loire		
Indre-et-Loire III	888-898	Coupland (2020)	Indre-et-Loire		
Isle-Aumont I	814-840	Haertle (1997)	Isle-Aumont	48.21131	4.12459
Isle-Aumont II	864-898	Haertle (1997)	Isle-Aumont	48.21131	4.12459
Issy l'Evêque	843-922	Morrison and Grunthal (1967)	Issy l'Evêque	46.70818	3.9734
Jedomelice	814-840	Coupland (2020)	Jedomelice	50.23411	13.971234
Jelsum	768-814	Morrison and Grunthal (1967)	Jelsum	53.23455	5.783862
Juaye-Mondaye	800-922	Morrison and Grunthal (1967)	Juaye-Mondaye	49.20803	-0.68508
Jura	768-814	Morrison and Grunthal (1967)	Jura		
Karden	814-822	Morrison and Grunthal (1967)	Karden	50.179051	7.299583
Karos-Eperjesszög I	888-915	Révész (1996)	Karos	48.32959	21.73712
Karos-Eperjesszög II	900-911	Gedai (1993)	Karos	48.32959	21.73712
Kättilstorp	814-877	Morrison and Grunthal (1967)	Kättilstorp	58.041694	13.711198
Katwijk I	800-922	Kluge (1993)	Katwijk	52.195273	4.421091
Katwijk II	794-800	Velde (2008)	Katwijk	52.195273	4.421091
Kecel	888-924	Huszár (1955)	Kecel	46.52644	19.24647
Kenézlő	826-950	Huszár (1955)	Kenézlő	48.2	21.53333
Kimswerd-Pingjum I	814-877	Morrison and Grunthal (1967)	Kimswerd	53.1289	5.4387
Kimswerd-Pingjum II	814-878	Morrison and Grunthal (1967)	Kimswerd	53.1289	5.4387
Kiskundorozsma-Hosszúhát	826-950	Múzeum Móra Ferenc (2002)	Szeged	46.275	20.06278
Kiskunfélegyháza	881-918	Kovács (1989)	Kiskunfelegyhaza	46.71246	19.85279
Koblenz	823-830	Reinhold Fischer Auktionshaus (2010)	Koblenz	50.359618	7.59383
Krinkberg	768-814	Morrison and Grunthal (1967)	Pöschendorf	54.03055	9.472156
La Cornouaille	814-877	Coupland (2020)	La Cornouaille	47.578279	-0.797543
La Couvertoirade	881-898	Coupland (2011a)	La Couvertoirade	43.91127	3.31355
La Roche en Ardenne	750-950	Coupland (2014)	La-Roche-en-Ardenne	50.183528	5.575243
La Tessoualle	814-877	Haertle (1997)	La Tessoualle	47.00535	-0.8494
La Tour-de-Peilz	755-768	Geiser (1990)	La-Tour-de-Peilz	46.45302	6.85686
Ladánybene	888-922	Huszár (1955)	Ladánybene	47.03333	19.45
Lamairé	843-877	Baigl, Clairand, and Jeanne-Rose (1995)	Lamairé	46.75707	-0.1263
Lamotte Beuvron	814-877	Coupland (2020)	Lamotte-Beuvron	47.602363	2.025245
Langon	814-877	Morrison and Grunthal (1967)	Langon	44.55389	-0.24833

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Hoard Name	Date	Reference	Location	Latitude	Longitude
Langres I	843-922	Morrison and Grunthal (1967)	Langres	47.85816	5.33113
Langres II	864-884	Coupland (2011a)	Langres	47.85816	5.33113
Larino	768-840	De Benedittis and Lafaurie (1998)	Larino	41.7968	14.9128
Lauterach	840-924	Zäch and Tabernero (2002)	Lauterach	47.4745	9.730031
Lauzès	814-877	Morrison and Grunthal (1967)	Lauzès	47.4707	-0.55324
Lavelanet	888-898	Coupland (2020)	Lavelanet	42.932652	1.848583
Laxfield	843-877	Morrison and Grunthal (1967)	Laxfield	52.30114	1.36237
Leiderdorp	768-840	Coupland (2020)	Leiderdorp	52.151653	4.529015
Lésigny-sur-Creuse	814-898	Jeanne-Rose (1996)	Lésigny-sur-Creuse	46.84996	0.76421
Levice-Géňa	926-950	Minarovicova (2007)	Levice-Géňa	48.21639	18.60806
Lillebonne	814-877	Coupland and Moesgaard (2012)	Lillebonne	49.51802	0.53681
Limoux	849-877	Haertle (1997)	Limoux	43.053658	2.217421
Lisówek	848-922	Morrison and Grunthal (1967)	Lisówek	51.9	20.9333
Llanbedrgoch	814-878	Coupland (2020)	Llanbedrgoch	53.300117	-4.236622
Llerida	887-928	Doménech-Belda and others (2013)	Lleida	41.61879	0.621737
Loire River Bank	814-840	Coupland (2014)	Loire River		
Loiret	843-1027	Duplessy (1985)	Loiret		
Lokeren	843-864	Haertle (1997)	Lokeren	51.10473	3.9865
Longjumeau	843-884	Moesgaard (2010)	Longjumeau	48.69173	2.29005
Loppersum	814-877	Morrison and Grunthal (1967)	Loppersum	53.33276	6.74398
Lucca	947-961	Saccocci and others (2004)	Lucca	43.84201	10.51534
Lussac-les-Châteaux	845-848	Haertle (1997)	Lussac-les-Châteaux	46.403093	0.723563
Lutkesaaxum	843-864	Haertle (1997)	Lutkesaaxum	53.364638	6.489072
Luzancy	814-877	Sombart (2008)	Luzancy	48.97205	3.1865
Lyon	751-771	Coupland (2020)	Lyon	45.758973	4.830895
Maine et Loire	751-878	Coupland (2014)	Maine-et-Loire		
Marçay	840-898	Morrison and Grunthal (1967)	Marçay	47.10002	0.21706
Marssum	814-855	Coupland (2011a)	Marssum	53.21056	5.73008
Marsum	814-887	Morrison and Grunthal (1967)	Marsum	53.339476	5.73008
Matha	778-877	Coupland (2014)	Matha	45.867625	-0.321187
Melle I	875-877	Haertle (1997)	Melle	46.221471	-0.147358
Melle II	843-877	Haertle (1997)	Melle	46.221471	-0.147358
Melle IV	823-825	Coupland (2018)	Melle	46.221471	-0.147358
Mercurey	822-877	Duplessy (1985) and Haertle (1997)	Mercurey	46.833364	4.722119
Méréville	814-877	Morrison and Grunthal (1967)	Méréville-Saint-Pierre	48.59069	6.15058
Metz	843-877	Morrison and Grunthal (1967)	Metz	49.11566	6.1732
Meurthe et Moselle	898-922	Coupland (2014)	Meurthe-et-Moselle		
Midlaren	814-877	Morrison and Grunthal (1967), Haertle (1997)	Midlaren	53.1111	6.67616
Midlum	900-961	Morrison and Grunthal (1967)	Midlum	53.18204	5.44716
Mikulčice	887-900	Slovenská akadémia vied. Archeologickỳ ústav (1979)	Mikulčice	48.81667	17.05
Molliens-Vidame	817-877	Haertle (1997)	Molliens-Dreuil	49.8839	2.02
Monchy-au-Bois	840-922	Morrison and Grunthal (1967)	Monchy-au-Bois	50.17999505	2.656698281
Montmain	768-814	Coupland (2020)	Montmain	49.410716	1.252625
Montrieux-en-Sologne II	800-922	Morrison and Grunthal (1967)	Montrieux-en-Sologne	47.55408	1.72638
Montrieux-en-Sologne III	864-898	Morrison and Grunthal (1967)	Montrieux-en-Sologne	47.55408	1.72638
Moreria		Parvérie (2018)	Moreria	38.916776	-6.349645
Mourlieu	900-925	Caron (1882)	Mourlieu	46.564931	0.512703
Muizen	822-877	Morrison and Grunthal (1967)	Muizen	51.01056	4.514722
Mullaghboden	814-877	Morrison and Grunthal (1967)	Mullaghboy	54.83536	-5.72671
Muret	814-840	Coupland (2020)	Muret	43.460924	1.327252
Nagyszokoly	926-947	Kovács (1989)	Nagyszokoly	46.72132	18.21182
Nagyvázsony	902-947	Kovács (1989)	Nagyvázsony	46.9835	17.69408
Neufchateau I	800-922	Coupland (2014)	Neufchateau	48.356071	5.692627
Neufchateau II	814-848	Coupland (2014)	Neufchateau	48.356071	5.692627
Neuvy-au-Houlme	814-877	Morrison and Grunthal (1967), Duplessy (1985)	Neuvy-au-Houlme	48.8181	-0.19966
Niederlahnstein	855-869	Coupland (2020)	Niederlahnstein	50.315193	7.598382
Nourray	843-877	Morrison and Grunthal (1967)	Nourray	47.71903	1.06023
Nr.Trier	768-855	Coupland (2014) and Morrison and Grunthal (1967)	Trier	49.755513	6.640075
Orléans	814-864	Haertle (1997)	Orléans	47.90143	1.90496
Oudwoude	814-877	Morrison and Grunthal (1967)	Oudwoude	53.27968	6.11413
Palma de Majorque	800-888	Doménech-Belda and others (2013)	Palma de Majorque	39.570589	2.648991
Paule	843-877	Coupland (2014)	Paule	48.235953	-3.444348
Pilligerheck	814-877	Petry and Wittenbrink (2021), Coupland (2011b)	Muenstermaifeld	50.20461	7.31152
Pingjum	900-911	Morrison and Grunthal (1967)	Pingjum	53.11519	5.44004

Continued on next page

Hoard Name	Date	Reference	Location	Latitude	Longitude
Place Unknown	954-986	Morrison and Grunthal (1967)			
Plessé	875-877	Haertle (1997)	Plessé	47.54109	-1.88812
Poitou Charentes	814-877	Coupland (2020)	Poitou-Charente		
Pommern	887-924	Coupland (2020)	Pommern	50.169368	7.269726
Pont Saint-Pierre	864-877	Coupland (2011a)	Pont-Saint-Pierre	49.33388	1.2745
Postsaal	814-1024	Coupland (2020)	Bavière		
Pouzauges	875-898	Haertle (1997)	Pouzauges	46.7822	-0.8361
Questembert	814-877	Haertle (1997)	Questembert	47.66097	-2.4521
Raalte	814-877	Coupland (2011a)	Raalte	52.38724	6.27462
Regensburg	843-877	Haertle (1997)	Regensburg	49.016213	12.097468
Rennes	843-922	Morrison and Grunthal (1967)	Rennes	48.10761	-1.68448
Rijs	814-877	Morrison and Grunthal (1967)	Rijs	52.86298	5.49838
Rijswijk	814-840	Coupland (2020)	Rijswijk	52.039942	4.325633
Rochefort	900-911	Coupland (2020)	Rochefort	45.935077	-0.962458
Roches l'Evêque	814-922	Morrison and Grunthal (1967)	Roches l'Evêque	47.7772	0.8922
Roermond	796-877	Haertle (1997), Coupland (2011b), Zuyderwyk and Besteman (2010)	Roermond	51.193179	5.98624
Rome I (Forum)	887-950	Metcalf (1992)	Rome	41.90509	12.46194
Rome II (Vatican)	898-922	Morrison and Grunthal (1967)	Rome	41.90509	12.46194
Rosas	814-840	Doménech-Belda and others (2013)	Rosas	42.265002	3.178593
Roswinkel	768-882	Morrison and Grunthal (1967)	Roswinkel	52.83787	7.03843
Rotterdam	814-840	Coupland (2020)	Rotterdam	51.919909	4.47544
Saint Bris le Vineux	814-877	Coupland (2020)	Saint-Bris-le-Vineux	47.74291	3.651349
Saint Ponc	884-887	Doménech-Belda and others (2013)	Saint-Ponç	41.963245	1.603627
Saint Yrieix la Perche	888-898	Coupland (2020)	Saint-Yrieix-la-Perche	45.51359	1.203618
Saint-Brieuc	864-875	Haertle (1997)	Saint-Brieuc	48.5136	-2.7653
Saint-Calais	768-877	Paty (1848)	Saint-Calais	47.9211	0.7439
Saint-Cyr-en-Talmondais	814-877	Morrison and Grunthal (1967)	Saint-Cyr-en-Talmondais	46.4614	-1.3356
Saint-Denis	793-875	Haertle (1997)	Saint-Denis	48.9364	2.3547
Saint-Martin-sur-le-Pré		Coupland (2014)	Saint-Martin-sur-le-Pré	48.9778	4.3394
Saint-Même-le-Tenu	814-877	Coupland (2014)	Saint-Même-le-Tenu	47.020808	-1.794104
Saint-Michel-de-Chavaignes		Haertle (1997)	Saint-Michel-de-Chavaignes	48.018584	0.570918
Saint-Pierre-de-Maillé	814-840	Benoit and Braunstein (1983)	Saint-Pierre-de-Maillé	46.6797	0.8444
Saint-Pierre-des-Fleurs I	823-877	Coupland and Moesgaard (2012)	Saint-Pierre-des-Fleurs	49.2514	0.9667
Saint-Pierre-des-Fleurs II	888-898	Cardon, Moesgaard, Prot, and Schiesser (2008)	Saint-Pierre-des-Fleurs	49.2514	0.9667
Saint-Seine-l'Abbaye		Coupland (2014)	Saint-Seine-l'Abbaye	47.440003	4.788637
Santa Elena	961-966	Doménech-Belda and others (2013)	Irun	43.337137	-1.786251
Santiago de Compostela	800-888	Doménech-Belda and others (2013)	Santiago de Compostela	42.880265	-8.543118
Sarlat	814-877	Coupland (2020)	Sarlat-la-Canéda	44.889865	1.216381
Sarzana	768-814	Morrison and Grunthal (1967)	Sarzana	44.11186	9.95886
Saumeray	843-877	Morrison and Grunthal (1967)	Saumeray	48.25027	1.32157
Saumur-Thouars	843-898	Morrison and Grunthal (1967)	Saumur	47.1218	-0.1704
Saverne		Duplessy (1985)	Saverne	48.73947	7.36602
Savigné-sous-le-Lude	843-898	Morrison and Grunthal (1967)	Savigné-sous-le-Lude	47.61845	0.05801
Savigny en Véron	814-877	Coupland (2020)	Savigny-en-Véron	47.205554	0.147106
Seiches sur le Loir	751-814	Coupland (2014)	Seiches-sur-le-Loir	47.578315	0.362977
Séranon	814-840	Coupland (2020)	Séranon	43.772823	6.704362
Sevilla region	888-898	Parvérie (2018)	Sevilla	37.393305	-5.993535
's-Hertogenbosch	814-840	Coupland (2014)	's-Hertogenbosch	51.698578	5.303773
Sigean	768-814	Coupland (2020)	Sigean	43.0287	2.978539
Silverdale	800-898	Coupland (2014)	Silverdale	54.167322	-2.82505
Minor Finds	751-1027	Morrison and Grunthal (1967)			
Søndre Bø	814-883	Morrison and Grunthal (1967)	Søndre Bø	58.11019	6.88224
Strasbourg-Basel	843-954	Morrison and Grunthal (1967)	Strasbourg/Basel	48.171	7.6473
Szabadbattyán	826-950	Huszár (1955)	Szabadbattyán	47.11798	18.3629
Szabadegyháza	888-924	Kovács (1989)	Szabadegyháza	47.07845	18.69228
Szedeg-othalom	902-924	Coupland (2014)	Szeged	46.265179	20.140614
Szekszárd	902-947	Huszár (1955)	Szekszárd	46.34779	18.70626
Tarrega	887-928	Doménech-Belda and others (2013)	Tarrega	41.648564	1.140707
Taizy	864-877	Coupland (2020)	Taizy	49.51967	4.25832
Teloché	864-877	Hucher (1845)	Teloché	47.88987	0.26731
Ter Apel	900-911	Morrison and Grunthal (1967)	Ter Apel	52.878359	7.063981
Ter Heijde	814-840	Coupland (2020)	Ter Heijde	52.02903	4.164265
Terslev	814-966	Morrison and Grunthal (1967)	Terslev	55.37476	11.9693

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Hoard Name	Date	Reference	Location	Latitude	Longitude
Thoiry	875-894	Haertle (1997)	Thoiry	48.86519	1.79463
Thouars	822-855	Morrison and Grunthal (1967)	Thouars	46.977604	-0.21579
Tiel	898-922	Coupland (2011a)	Tiel	51.88809	5.43069
Tiszaeszlàr I	814-950	Kovács (1989)	Tiszaeszlàr	48.05	21.46667
Tiszaeszlàr II	926-950	Kovács (1989)	Tiszaeszlàr	48.05	21.46667
Tiszanána	888-946	Kovács (1989)	Tiszanána	47.56111	20.52382
Troyes	814-840	Coupland (2014)	Troyes	48.299055	4.077872
Troyes II	843-877	Coupland (2020)	Troyes	48.58345	3.81356
Tuscany	888-973	Ciampoltrini, Abela, and Bianchini (2001)	Tuscany		
Tytsjerksteradiel	814-855	Coupland (2020)	Burgum	53.195748	5.987155
Tzummarum I	819-855	Haertle (1997)	Tzummarum	53.238297	5.549116
Tzummarum II	855-865	Coupland (2020)	Tzummarum	53.238297	5.549116
Unknown	954-986	Morrison and Grunthal (1967)	France		
Vale of York	898-922	Williams and Ager (2010)	Vale of York	54.20361	-1.36398
Valence	819-840	Haertle (1997)	Valence	44.93347	4.890808
Vallée de la Risle	814-877	Coupland and Moesgaard (2012)	Vale of Risle	49.424	0.725
Vercelli	768-814	Morrison and Grunthal (1967)	Vercelli	45.32255	8.41844
Verdun I	875-877	Haertle (1997)	Verdun	49.15952	5.382316
Verdun II	881-887	Morrison and Grunthal (1967)	Verdun	49.15952	5.382316
Vereb	858-924	Morrison and Grunthal (1967)	Vereb	47.31867	18.61802
Vernon	814-877	Coupland (2020)	Vernon	49.091052	1.483426
Vicq sur Gartempe	814-877	Coupland (2020)	Vicq sur Gartempe	46.721302	0.862012
Vire	843-877	Morrison and Grunthal (1967)	Vire-Normandie	48.83919	-0.89
Vrigny	843-877	Haertle (1997)	Vrigny	48.08167	2.243889
Wagenborgen	814-877	Haertle (1997)	Wagenborgen	53.25713	6.93525
Westerklief I	814-877	Sarfatij, Verwers, and Woltering (1999)	Westerklief	52.89494	4.93322
Westerklief II	814-877	Besteman (2006)	Westerklief	52.89494	4.93322
Wiesbaden-Biebrich	717-814	Morrison and Grunthal (1967)	Wiesbaden-Biebrich	50.050115	8.237668
Wijk bij Duurstede I	793-822	Morrison and Grunthal (1967)	Wijk-Bij-Duurstede	51.971869	5.344562
Wijk bij Duurstede II	752-768	Van Es and Verwers (1980)	Wijk-Bij-Duurstede	51.971869	5.344562
Wijk bij Duurstede III	768-820	Van Es and Verwers (1980)	Wijk-Bij-Duurstede	51.971869	5.344562
Wijk bij Duurstede IV	823-840	Dijkstra (2005)	Wijk-Bij-Duurstede	51.971869	5.344562
Wijk bij Duurstede V	751-768	Coupland (2020)	Wijk-Bij-Duurstede	51.971869	5.344562
Wirdum	814-877	Coupland (2020)	Wirdum	53.149585	5.803308
Worms	814-840	Coupland (2020)	Worms	49.632241	8.36221
Yde	814-877	Morrison and Grunthal (1967)	Yde	53.11143	6.58365
Yonne	814-840	Coupland (2014)	Yonne	47.89753	3.588695
York	751-887	Dolley (1965)	York	53.95333	-1.08342
Yronde	843-877	Morrison and Grunthal (1967)	Yronde	45.6133	3.25481
Zelzate	814-877	Morrison and Grunthal (1967)	Zelzate	51.19753	3.81463
Zetel	768-793	Völckers (1965)	Zetel	53.4146	7.9699
Zillis	888-949	Zäch (2001)	Zillis	46.6355	9.44514
Zuidlaren	875-894	Haertle (1997)	Zuidlaren	53.09231	6.679414

TABLE D.V: Manual mint codings I: new mints

Mint	id	Location	Lat.	Long.	Notes
Abarqubadh	abarqubadh		31.28027	47.49266	"This mint was in the district of Khusra-shadh Bahmân (the district of the Tigris) in Irâq, between Wâsit and al-Basra and near the border with Khuzistân." Lloyd (2023)
Adurbadagan	adurbadagan	Ganzak	37.0123	46.2019	Sasanian mint (AT)
al Hashimiyah	al-hashimiyah	Kufa	32.05114	44.44017	Rare Abbasid mint during al-Mansur's reign, situated close to Kufah (138-146 AH).
al Rahba	al-rahba	Mayadin	35.005	40.4235	A mint in Syria, on the Euphrates
Arrajan	arrajan		30.65388	50.27472	"[Bizamqubadh] was an alternative name for Arrajân in Fars, and also appears to have struck Arab-Sasanian issues." Lloyd (2023)

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Mint	id	Location	Lat.	Long.	Notes
Hulwan	hulwan		34.465	45.855	"This mint-name is that of a district (astân) in Irâq, which covered an area to the north-east of Baghdad. Le Strange notes that this district was also known as Shâd Fîrûz - presumably its former Sasanian name. The town of Hulwân itself evidently lay just over the border in Jibâl province, although at this period it appears to have been included with 'Irâq for administrative purposes." Lloyd (2023)
Madinat Elvira	madinat_elvira		37.23105	-3.70848	The archaeological site of Madinat Ilbira.
Mah al Basrah	mah-al-basrah	Nihavand, Iran	34.18879	48.37046	"The term is the Arabic name for Nihavand." (British Museum `x107840`)
Mah al Kufah	mah-al-kufah	Dinawar, Iran	34.583333	47.43333	"Mah al-Kufah = Dinawar (sometimes incorrectly written Daynawar) in the middle ages was one of the most important towns in Djibal (Media); it is now in ruins. The exact location is 34 degrees 35 minutes Lat. N. and 47 degrees 26 minutes E. Long. (Greenwich)." Lockhart (2012)
Masabadhan	masabadhan		33.52303	46.86539	A district with capital al-Sirawan; the location of al-Sirawan is from Cornu (1983)'s atlas.
Maysan	maysan	Naysan	30.8093	47.5628	Sasanian mint Maysan (MY)
Panjshir	panjshir	Panjshir Valley	35.254095	69.456014	Panjshir Valley, modern-day Afghanistan
Rev-Ardashir	rev-ardashir	Bushehr	28.9119	50.8367	Sasanian mint (LYW/ LYWARTHST/KWN LYW/GNC LYW); the location is from FLAME
Roda	roda		42.26478	3.17887	A carolingian mint in Rosas, Spain
Sarakhs	sarakhs	Sarakhs, Iran	36.5449	61.1577	"A town in Khurâsân located roughly midway between Marw and Abrashahr. Sarakhs lay on the eastern bank of the Mashhad river, about forty or fifty miles north of its confluence with the Herât river." Lloyd (2023)
Uman	oman	Oman	23.51234	58.27000	Lloyd (2023): "Modern Oman on the Persian Gulf."

TABLE D.VI: Manual mint codings II: geocoding existing Nomisma mints

Nomisma ID	Nomisma Note	Lat.	Long.	Note
al-Abbasiyah	"Earfly Abbasid site in North Africa"	35.62183407	10.18089991	According to Abdul Wahab (2012), three miles south-east of Kairouan.
al-Furat	"In the district of Shadh Bahman in Iraq, but its exact location unknown. Klat, 16."	30.53269083	47.87593421	Geocodes based on Fig. 11 in Morony (1982).
al-Madinat al-Mutawakkiliyah	"al-Madinat al-Mutawakkiliyah is just north of Sammara and was built by the Abbasids."	34.2621862	43.85500034	Close to Samarra, Iraq
al-Manadhir	"The name of two districts, with tehir chief-towns, named Greater Manadhir & Little Manadhir in Khuzistan, Iran"	31.97753445	48.69644554	Lloyd (2023): "Manâdhir was a district within the province of Khuzistân, situated between the Dizfûl and Dujayl rivers above their confluence north of Ahwâz. It was apparently divided into two parts named Greater and Little Manâdhir, each containing a chief town with the same name."
al-Mubarakah (Abbasid)	"Some place in North Africa"	36.30565739	10.13850323	Unknown location, coding it to modern-day Tunisia
al-Samiyah	"Al-Samiya was in the Shatt al-Arab area of lower Iraq."	30.6617666	47.78548511	Coding to Shatt al-Arab.
Bihqubadh af-Asfal	"Lower Bihqubach (sic) in Iraq on the Euphrates"	31.56718959	45.22725183	Lloyd (2023): "The three districts of Upper, Middle and Lower Bihqubâdh were located in 'Iraq to the west of the Euphrates. Bihqubâdh is taken from the Persian meaning "the good land of king Qubâdh. Al-Asfal means 'the lower,' and covered the land next to the Euphrates where it entered the Great Swamp." Coordinates based on Fig. 8 of Morony (1982).

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Nomisma ID	Nomisma Note	Lat.	Long.	Note
Dastawa	"South of Qazvin"	35.75554989	50.08839336	
Ma'din Bajunays	"Province north of Lake Van"	40.223509	43.8355181	Location very approximate in western Armenia.
Mani	"Klat is uncertain of its location although the prefix Mah occurs in older names for Dinavar & Nihavand. Quarter of Jibal. Klat"	34.38582341	47.97904114	Lloyd (2023) puts it either at Mah al Basrah (Nihavand) or Mah al Kufa (Dinavar). Our chosen geocode is halfway between the two.
Nahr Tira	"Exact location on the river or canal of the same name in Khuzistan not know. Klat, p.. 18"	30.8755	49.7131	From FLAME.
Qumis	"A small province which stretches along the foot of the Great Alburz chain of mouintains. Klat, p. 17."	35.96088616	54.03571139	Wikipedia "Qumis (region)"
Surraq	"Surraq or DAWRAQ (or Dawraq al-Fors), name of a district (kūra; Moqaddasī, pp. 406-07), also known as Sorraq, and of a town that was sometimes its chef-lieu in medieval Islamic times."	30.65094882	48.67463446	Coding to Shadegan, Iran.
Tabaristan	"Tabaristan, also known as Tapuria, was the name of the former historic region in the southern coasts of Caspian Sea roughly in the location of the northern and southern slopes of Elburz range in Iran."	36.5656	53.0588	From FLAME
Tudghah	"Unknown location in Morroco"	31.523	-5.5313	'Ush (1982) identifies it with "Todr'a", and cites Renou (1846) (incorrectly as authored by "Lavoix") saying that it was located fourty kilometers west of Sijilmasa, at a river of the same name. That would place it close to Tinghir, Morocco.

TABLE D.VII: Sasanian mint codings

Signatures	Approx. Location Nomisma ID	Notes
AHM	hamadhan	
AIRAN, AYLAN	hulwan	Eran-asankar-Kavad
AM	amol	Amol, Khorasan
APL, APR	nishapur	
ART, TART	ardashir_khurrah	TART: Tawwaj as dependency of Ardashir Khurra
AT	adurbadagan	
AU, AW	suq_al-ahwaz	AU is used by Al-Ush, we interpret it as "AW", Hormizd-Ardashir
AY, AYL	al-sus	Eran-khvarrah-Shapur. AYL: British Museum says "possibly referring to Susa."
BBA	ctesiphon	Court mint, probably at Ctesiphon (Gyselen)
BCLA, BJRA, DS, DST	al-basrah	Mallon-McCorgray interprets BCLA as al-Basra. Accoring to Schindel (2005) BJRA is al-Basra.
BISH, BYS, BYSH	bishapur	
BN, BRMKRMAN, DL, DR, GLM, KL, KLMAN, KLMANLCN, KR, KRAMAN H P, KRMAN, KRMAN W ST, KRMAN-GY, KRMAN-NAR, KRMAN-NAW, NAL, NAR	kirman	Multiple mints that are in Kirman province.
D', DA, DAP	darabjird	
DAP	fasa	

Continued on next page

Signatures	Approx. Location Nomisma ID	Notes
GD	jayy	
GU, GW	gorgan	We follow Schindel (2005) in attributing GW to Gorgan (after Yazdegerd I). Gyselen (1977) attributes GU to Gorgan.
HL	harat	
HWC	jundi_sabur	
LAM, RAM	ramhurmuz	
LD, RD	rayy	
LYW, RIU	rev-ardashir	Bivar (1970) associates RIU with LYW, and confirms Nö's interpretation as Rev-Ardashir
MA	masabadhan	
MB, MY, PL	maysan	
ML, MR	marw	
NH, NIHJ, NYHC, WH, WYHC	ctesiphon*	NH, WH: Veh-Ardashir. On WYHC, Album (2011): "A mint in northern Iraq, ostensibly the treasury mint near Ktesiphon prior to the AH50s, and thereafter, for a series dated AH67-73, Arrajan". We follow Album (2011), Schindel (2005), and others in attributing it to Ctesiphon before AH50, then Arrajan.
NHR	nahr_tira	
NIH, WYH	bihqubadh_af-asfal	
NIHJ	arrajan	Almost certainly the same as WYHC.
NY, NYH	antiocheia_persis	NY: Nihawand. For NYH, Schindel (2005) suggests Nihawand.
SHI	shiraz	
SK	zaranj, sijistan	
ST	istakhr	
SY	fars_shiraz	Unlocated mint, probably in Fars province (or Kirman, as has sometimes been suggested).
TPWRSTAN	tabaristan	
YZ, ZR, GZ	yazd	

E. IDENTIFICATION

E.1. *Asymptotic identification proof*

Our identification proof proceeds in three steps.

First (section E.1.1), we show that under our random sampling assumption (18) for the data generating process (see appendix tables B.VI and B.VII for empirical evidence in support of this assumption), our maximum likelihood estimator exactly recovers the true shares of coins within the coin stock of a single region $\times$ period from data on a single hoard for that region $\times$ period. Second (section E.1.2), we extend this result to all hoards and show that arbitrarily normalizing the total hoard sizes has no impact on the estimation of true shares of coins within hoards. Third (section E.1.3), we show constructively how to uniquely recover all parameters of the true model (up to the normalizations defined in section 4) from knowledge of shares of coins within hoards.

E.1.1. *Asymptotic recovery of coin shares by maximum likelihood, single hoard*

Under our data generating process assumption in equation (18), i.e. hoards in our dataset are random samples from the coins in circulation, sample shares (within a single hoard) are asymptotically equal to model predicted shares. Let define $\mathcal{N}$, the number of coins found in a given (h, T) hoard buried in region h at time T (*tpq*). From the law of large numbers,

$$\lim_{\mathcal{N}\to+\infty} \frac{H_{m,h}[t,T]}{H_h[T]} = \mathbb{E}\left[\frac{H_{m,h}[t,T]}{H_h[T]}\right] = \frac{S_{m,h}[t,T]\left(\boldsymbol{\Theta}^{\text{true}}\right)}{\sum\limits_{m',t'\leq T} S_{m',h}[t',T]\left(\boldsymbol{\Theta}^{\text{true}}\right)} \equiv \left(H(t,T)\right)^{\text{true}}_{m,h}.$$

Observed coin shares within hoards $\left({}^{H_{m,h}[t,T]}/{}_{H_h[T]}\right)$ are asymptotically equal to model-predicted true shares within coin stocks $\left({}^{S_{m,h}[t,T]\left(\boldsymbol{\Theta}^{\text{true}}\right)}/{}_{\sum_{m',t'\leq T} S_{m',h}[t',T]\left(\boldsymbol{\Theta}^{\text{true}}\right)}\right)$, which we label $(H(t,T))^{\text{true}}_{m,h}$.

The next step is to prove, for a single hoard (h,T), that the likelihood of observing coin shares is (asymptotically) maximized when estimated shares, $(H(t,T))^{\text{est.}}_{m,h}$ are equal to realized shares $(H(t,T))^{\text{true}}_{m,h}$. Section E.1.2 trivially extends this result for multiple hoards.

Arbitrarily choosing one reference region $\times$ period (m_0,t_0) as shares add up to one,

$$\arg\max_{\left(\cdots,(H(t,T))^{\text{est.}}_{m,h},\cdots\right)_{(m,t)\neq(m_0,t_0)}} \left\{ \begin{aligned} \mathcal{L} = &\left(\sum_{m,t}(H(t,T))^{\text{true}}_{m,h}\ln(H(t,T))^{\text{est.}}_{m,h}\right) \\ &+\left(1-\sum_{m',t'}(H(t',T))^{\text{true}}_{m',h}\right)\ln\left(1-\sum_{m'',t''}(H(t'',T))^{\text{est.}}_{m'',h}\right)\end{aligned}\right\}$$
$$=\left(\cdots,(H(t,T))^{\text{true}}_{m,h},\cdots\right)_{(m,t)\neq(m_0,t_0)}$$

PROOF: From the FOC's wrt each share $(H(t,T))^{\text{est.}}_{m,h}$:

$$\frac{(H(t,T))^{\text{true}}_{m,h}}{(H(t,T))^{\text{est.}}_{m,h}} - \frac{(1-\sum_{m',t'}(H(t',T))^{\text{true}}_{m',h})}{(1-\sum_{m',t'}(H(t',T))^{\text{est.}}_{m',h})} = 0, \forall m,t \Rightarrow (H(t,T))^{\text{est.}}_{m,h} = (H(t,T))^{\text{true}}_{m,h}, \forall m,t$$

Q.E.D.

E.1.2. *Irrelevance of coin hoard sizes for maximum likelihood, multiple hoards*

From section E.1.1, any strictly monotonically increasing transformation (e.g. logged and multiplied by a positive constant $\alpha(h,T)>0$) of the likelihood of observing a distribution of coin types within a single hoard is maximized for estimated shares equal to true shares,

$$\arg\max_{\left(\cdots,(H(t,T))^{\text{est.}}_{m,h},\cdots\right)_{(m,t)\neq(m_0,t_0)}} \alpha(h,T)\sum_{m,t\leq T}(H(t,T))^{\text{true}}_{m,h}\ln(H(t,T))^{\text{est.}}_{m,h}$$
$$=\left(\cdots,(H(t,T))^{\text{true}}_{m,h},\cdots\right)_{(m,t)}, \forall(h,T), \forall\alpha(h,T)>0,$$

because the scalar $\alpha(h,T)>0$ cancels out from the FOC's in E.1.1.

Adding up the log-likelihood for multiple hoards, we get,

$$\arg\max_{\left(\cdots,(H(t,T))^{\text{est.}}_{m,h},\cdots\right)_{(m,t)\neq(m_0,t_0)}}$$

$$\sum_{h,T} \alpha(h,T) \sum_{m,t\leq T} (H(t,T))^{\text{true}}_{m,h} \ln (H(t,T))^{\text{est.}}_{m,h}$$
$$= \left(\cdots, (H(t,T))^{\text{true}}_{m,h}, \cdots\right)_{(m,h,t,T)}, \forall\alpha(h,T) > 0.$$

The FOC's wrt coin shares for one hoard is independent from the FOC's for other hoards, so the system of FOC's for each (h,T) hoard implies $(H(t,T))^{\text{est.}}_{m,h} = (H(t,T))^{\text{true}}_{m,h}$, $\forall m,t$, $\forall \alpha(h,T) > 0$. In other words, any selection bias that increases the likelihood of hoards being buried, discovered, and documented does not affect our estimates.

E.1.3. *Unique recovery of parameters from coin shares within hoards*

We now proceed to the proof that knowledge of the (asymptotic) shares of coins within hoards is sufficient to uniquely recover the true structural parameters governing the flow of coins.[8] This proof considers a simplified setting with a single determinant of bilateral trade flows (travel times) and zero saving. Extending the proof to include political and religious borders is straightforward. Extending the proof to include positive saving is also straightforward but extremely tedious.

<u>*Problem Formulation*</u>

Latent Coin Dynamics. We define a dynamic system evolving over discrete periods $t = 1,\dots,T$. For any periods $t < s$, the coin stock matrix $S(t,s) \in \mathbb{R}^{N\times N}_{++}$ is defined as in (10):

$$S(t,s) = (1-\lambda)^{s-t} M(t)\Pi(t+1)\dots\Pi(s) \tag{E.1}$$

where:

- $\lambda \in (0,1)$ is a known coin loss rate.
- $M(t)$ is an unknown $N \times N$ invertible diagonal matrix of minting in period t.
- $\Pi(s)$ is an unknown $N \times N$ row-stochastic trade share matrix in period s.

Parametric Structure. The trade matrices $\Pi(s)$ follow a gravity structure governed by an iso-elastic function of travel times as in (11) and (15):

$$\Pi(s) = A(s)\Delta B(s) \tag{E.2}$$

- Δ is the matrix of bilateral trade terms with entries $\Delta_{ij} = d_{ij}^{-\zeta}$, with $D = [d_{ij}]$ a known matrix of bilateral travel times, and the elasticity ζ unknown.
- $B(s)$ is an unknown diagonal matrix of seller terms, $(B(s))_{ii} = \tilde{\beta}_i$. We impose the normalization constraint $(B(s))_{11} = 1$.
- $A(s)$ is the unknown diagonal normalization matrix of buyer terms, $A(s)_{ii} = \tilde{\alpha}_i$, defined uniquely by the fact that trade shares sum to one $\Pi(s)\mathbf{1} = \mathbf{1}$.

Observations. We observe the column-normalized matrices of coin shares within hoards, $H(t,s)$, with the share of coins minted in region m in period t among coins in a hoard buried in region h in period s corresponding to element $(H(t,s))_{m,h}$ defined as in section E.1.1.

[8] We used the support of Gemini 3.0 to derive this proof, in particular the formulation and solution of the fixed point problem in theorem 1 in step 2.

Sections E.1.1 and E.1.2 show that knowledge on empirical coin hoards is asymptotically sufficient to recover true shares of coins within coin stocks, which we now take as given: $(H(t,s))^{\text{true}} = (H(t,s))^{\text{est.}} = (H(t,s))$.

Let $\boldsymbol{\sigma}(s) = S(s)^T \mathbf{1}$ be the vector of column sums of the cumulative matrix of coin types in circulation at time s, $S(s) = \sum_{t=1}^{s-1} S(t,s)$. Let $\Sigma(s) = \text{diag}(\boldsymbol{\sigma}(s))$.

$$H(t,s) = S(t,s)\Sigma(s)^{-1} \tag{E.3}$$

Note that we only use information on shares of coins types within a hoard (summing up to one).

Goal. We aim to recover ζ (travel time elasticity), $M(t)$ (minting), and $B(t)$ (seller terms).

Step 1: Recovery of the elasticity parameter ζ

We first recover the elasticity ζ using ratios of observations. This is possible because the ratios collect the diagonal matrices $M(t)$, $A(t)$ and $B(t)$ into separate coin origin and destination fixed effects via the properties of logarithms.

Construct the observation ratio matrix Y between period 2 and period 3:

$$Y = H(1,2)^{-1}H(1,3) = (1-\lambda)\Sigma(2)\Pi(3)\Sigma(3)^{-1} \tag{E.4}$$

Substituting the structure of $\Pi(3)$:

$$Y_{ij} = (1-\lambda)\sigma_i(2)A_i(3)d_{ij}^{-\zeta}B_j(3)\sigma_j(3)^{-1} \tag{E.5}$$

Take the natural logarithm:

$$\ln(Y_{ij}) = \underbrace{\ln((1-\lambda)\sigma_i(2)A_i(3))}_{u_i} + \underbrace{\ln(B_j(3)\sigma_j(3)^{-1})}_{v_j} - \zeta\ln(d_{ij}) \tag{E.6}$$

We apply the Double Centering Operator (2-way fixed effect) $J = I - \frac{1}{N}\mathbb{J}$, where $\mathbb{J}$ is the matrix of ones, to annihilate the row and column terms:

$$J\ln(Y)J = -\zeta[J\ln(D)J] \tag{E.7}$$

The parameter ζ is uniquely recovered via Frobenius norm minimization:

$$\zeta = -\frac{\text{tr}((J\ln(D)J)^T(J\ln(Y)J))}{\text{tr}((J\ln(D)J)^T(J\ln(D)J))} \tag{E.8}$$

Step 2: Recovery of Relative Hoard Sizes via Fixed Point Theory

With ζ known, we compute the bilateral terms matrix $\hat{\Delta} = D^{-\zeta}$. We now seek to recover the mass vector $\boldsymbol{\sigma}(3)$ (number of coins within hoards) up to a global scalar κ.

Structural Decomposition. We remove the bilateral terms from the ratio matrix using Hadamard (element-wise) division:

$$Z = Y \oslash \hat{\Delta} \tag{E.9}$$

Analytically, $Z_{ij} = (1-\lambda)\sigma_i(2)A_i(3)B_j(3)\sigma_j(3)^{-1}$. This is a Rank-1 matrix. Compute the singular value decomposition: $Z = \mathbf{u}\mathbf{v}^T$. This establishes the functional dependency of the unknown $B(3)$ (seller terms) on the unknown $\boldsymbol{\sigma}(3)$:

$$\mathbf{v}_j \propto B_j(3)\sigma_j(3)^{-1} \implies B(\boldsymbol{\sigma}(3)) = \text{diag}(\mathbf{v} \circ \boldsymbol{\sigma}(3)) \tag{E.10}$$

Consequently, the normalization matrix $A(3)$ (buyer terms) is also a function of $\boldsymbol{\sigma}(3)$:

$$A(\boldsymbol{\sigma}(3)) = \text{diag}(\hat{\Delta}B(\boldsymbol{\sigma}(3))\mathbf{1})^{-1} \tag{E.11}$$

The Fixed Point Equation. We derive the coin mass conservation for $\boldsymbol{\sigma}(3)$.

1. **Coin flows from $t = 1$:** Since $Y = (1-\lambda)\Sigma(2)\Pi(3)\Sigma(3)^{-1}$, we have:

$$Y\boldsymbol{\sigma}(3) = (1-\lambda)\Sigma(2)\Pi(3)\mathbf{1} = (1-\lambda)\boldsymbol{\sigma}(2) \tag{E.12}$$

   Thus, the mass from the previous period is $\boldsymbol{\sigma}(2) = \frac{1}{1-\lambda}Y\boldsymbol{\sigma}(3)$.
2. **Minting injection at $t = 2$:** From $H(2,3)$, we have $(1-\lambda)M(2)\mathbf{1} = H(2,3)\boldsymbol{\sigma}(3)$.

Total Coin Mass Equation:

$$\boldsymbol{\sigma}(3) = \Pi(3)^T[(1-\lambda)\boldsymbol{\sigma}(2) + (1-\lambda)M(2)\mathbf{1}] \tag{E.13}$$

Substituting the components:

$$\boldsymbol{\sigma}(3) = \Pi(3)^T[Y\boldsymbol{\sigma}(3) + H(2,3)\boldsymbol{\sigma}(3)] \tag{E.14}$$

This yields the mapping $\mathcal{T} : \mathbb{R}^N_{++} \to \mathbb{R}^N_{++}$ for any candidate $\boldsymbol{\xi}$ for $\boldsymbol{\sigma}(3)$:

$$\boldsymbol{\xi} = \mathcal{T}(\boldsymbol{\xi}) \equiv \Pi(\boldsymbol{\xi})^T[Y + H(2,3)]\boldsymbol{\xi} \tag{E.15}$$

where $\Pi(\boldsymbol{\xi})$ is defined from (E.10) and (E.11) as $\Pi(\boldsymbol{\xi}) = A(\boldsymbol{\xi})\hat{\Delta}B(\boldsymbol{\xi})$.

Existence and Uniqueness Proof.

THEOREM 1: *The mapping $\mathcal{T}$ admits a unique solution in the projective space of positive rays.*

PROOF: We utilize the **Hilbert Projective Metric** $d_H(\mathbf{x}, \mathbf{y}) = \ln \frac{\max(x_i/y_i)}{\min(x_i/y_i)}$.

1. **Positivity:** Since d_{ij} is finite, $\hat{\Delta}$ is strictly positive. For $\boldsymbol{\xi} > 0$, $\Pi(\boldsymbol{\xi})$ is strictly positive. $K = Y + H(2,3)$ is non-negative and non-zero. Thus $\mathcal{T}$ maps the interior of the positive cone to itself.
2. **Homogeneity:** Consider a scalar $c > 0$.

$$\begin{aligned} B(c\boldsymbol{\xi}) &= cB(\boldsymbol{\xi}) \\ A(c\boldsymbol{\xi}) &= \text{diag}(\hat{\Delta}(cB(\boldsymbol{\xi})\mathbf{1}))^{-1} = c^{-1}A(\boldsymbol{\xi}) \\ \Pi(c\boldsymbol{\xi}) &= (c^{-1}A)\hat{\Delta}(cB) = \Pi(\boldsymbol{\xi}) \quad \text{(Scale Invariant)} \end{aligned}$$

   Therefore, $\mathcal{T}(c\boldsymbol{\xi}) = \Pi(\boldsymbol{\xi})^T K(c\boldsymbol{\xi}) = c\mathcal{T}(\boldsymbol{\xi})$. The map is homogeneous of degree 1.
3. **Contraction:** By Birkhoff's Theorem (Birkhoff, 1957, 1965), any strictly positive, homogeneous map is a strict contraction under d_H.

By the Banach Fixed Point theorem, there exists a unique fixed point ray. Let $\boldsymbol{\xi}^*$ be the unit-norm solution. The true mass vector is $\boldsymbol{\sigma}(3) = \kappa\boldsymbol{\xi}^*$, where κ is the **unknown global scaling factor for minting** (the reason why we need to normalize minting in one, single, region $\times$ period). *Q.E.D.*

Step 3: Exact Recovery of $B(3)$ and Scaled Recovery of $M(1)$ and $M(2)$

Exact Recovery of $B(3)$. Using the recovered direction $\boldsymbol{\xi}^*$, we calculate the candidate buyer terms diagonal:

$$\tilde{\mathbf{b}} = \mathbf{v} \circ \boldsymbol{\xi}^* \tag{E.16}$$

From Eq. (E.10), the true diagonal is $\mathbf{b}(3) \propto \tilde{\mathbf{b}}$. Unlike the mass vector, $B(3)$ is constrained.

$$(B(3))_{11} = 1 \tag{E.17}$$

This allows us to solve for the exact proportionality constant. Let $\alpha = 1/(\tilde{b})_1$.

$$B(3) = \alpha \cdot \mathrm{diag}(\tilde{\mathbf{b}}) \tag{E.18}$$

This matrix of seller terms is **exactly recovered** (no κ ambiguity), under the normalization of the seller term for region 1. Consequently, $A(3)$ and $\Pi(3)$ are exactly recovered.

Explicit Recovery of $M(1)$ and $M(2)$. The minting matrices $M(t)$ can only be recovered up to the global scalar κ. From the row sums of the unnormalized observations:

$$M(2) = \frac{1}{1-\lambda}\mathrm{diag}(H(2,3)\boldsymbol{\sigma}(3)) = \kappa\left[\frac{1}{1-\lambda}\mathrm{diag}(H(2,3)\boldsymbol{\xi}^*)\right] \tag{E.19}$$

$$M(1) = \frac{1}{1-\lambda}\mathrm{diag}(H(1,2)Y\boldsymbol{\sigma}(3)) = \kappa\left[\frac{1}{1-\lambda}\mathrm{diag}(H(1,2)Y\boldsymbol{\xi}^*)\right] \tag{E.20}$$

Step 4: Recursive Recovery of $M(s-1)$ and $B(s)$ for $s > 3$

Assume $\boldsymbol{\sigma}(s-1)$ is known (up to the scalar κ).

1. **Recover $\boldsymbol{\sigma}(s)$:** Summing the columns of the coin hoard observation $H(s-1,s)$,

$$(\boldsymbol{\sigma}(s) - \boldsymbol{\sigma}(s-1)) \oslash \boldsymbol{\sigma}(s) = H(s-1,s)^T\mathbf{1} \tag{E.21}$$

   Solve this linear system for $\boldsymbol{\sigma}(s)$.
2. **Recover $M(s-1)$:**

$$M(s-1) = \frac{1}{1-\lambda}\mathrm{diag}(H(s-1,s)^T\mathbf{1} \circ \boldsymbol{\sigma}(s)) \tag{E.22}$$

3. **Recover $B(s)$:** Isolate the trade share matrix:

$$\Pi(s) = \frac{1}{1-\lambda}M(s-1)^{-1}H(s-1,s)\Sigma(s) \tag{E.23}$$

   Compute $K_s = \Pi(s) \oslash \hat{\Delta}$. Singular value decomposition yields the right singular vector $\mathbf{q}$.

$$B(s) = \frac{1}{q_1}\mathrm{diag}(\mathbf{q}) \quad \text{(Exact Recovery)} \tag{E.24}$$

Step 5: Uniqueness of the Solution

We have established a constructive solution. We must now verify that no alternative sequence of matrices exists that generates the same data. The system possesses a gauge freedom if there exists a vector sequence $u(t)$ such that $u(t) = \Pi(t)u(t+1)$ with $u_1(t) = 1$ and $u(t) \neq \mathbf{1}$.

THEOREM 2: *If $N \leq T$ and the matrix D is generic, the solution is unique up to the global scalar κ.*

PROOF: Let $\mathbf{z} \in \mathbb{R}^N$ be a perturbation of the final gauge vector $u(T+1) = \mathbf{1} + \mathbf{z}$. For the normalization $(B)_{11} = 1$ to hold at all previous time steps, $\mathbf{z}$ must satisfy:

$$e_1^T \Pi(t)\Pi(t+1)\ldots\Pi(T)\mathbf{z} = 0 \quad \forall t \in \{1, \ldots, T\} \tag{E.25}$$

where $e_1 = [1, 0, \ldots, 0]^T$. Let $\mathcal{O}$ be the Observability Matrix where row k is $e_1^T \Pi(T-k+1)\ldots\Pi(T)$. Since the matrices $\Pi(t)$ are strictly positive (due to $A, B, D > 0$), they are irreducible. For generic parameters, the directions of the first rows of cumulative products span $\mathbb{R}^N$. Thus, $\text{rank}(\mathcal{O}) = N$. The only solution to $\mathcal{O}\mathbf{z} = \mathbf{0}$ is $\mathbf{z} = \mathbf{0}$. Therefore, $u(t) = \mathbf{1}$ is the only valid gauge, and the recovered parameter matrices are unique. *Q.E.D.*

E.2. *Numerical identification with sparse data, finite sample, and misspecified model*

See our working paper version Boehm and Chaney (2026) for additional details on our numerical exploration of identification and misspecification.

Identification with sparse data. We first show that our estimator is successful at recovering known (true) parameters even with extremely sparse data as in our millennia old dataset on coin hoards. We treat the estimated parameters from section 4, using the specification in column 2 of table III with multiple religious border effects, as known (true) parameters. We use those parameters to simulate the entire sequence of coin stocks and coin flows using our full structural model in section 3. We then collapse our simulated dataset of coin stocks into hoards of sizes normalized to 1 (so that we only have information on shares of coin types within each hoard). Finally, we estimate all parameters using this simulated dataset. We are able to recover, up to machine precision, the known (true) parameter vector $(\zeta, \boldsymbol{\kappa})$ governing bilateral trade costs. We are able to precisely recover true trade shares and true minting output, and equilibrium variables—nominal incomes, real consumption, and production capacity—with a high degree of accuracy. There are small discrepancy between known (true) values and estimated values due to numerical imprecisions.

Identification with random-sized hoards. We confirm our formal identification result that differences in hoard sizes have no impact on our ability to recover parameters. We start from the same estimated parameters, simulate our structural model, and simulate a coin hoard dataset, but instead of normalizing all hoard sizes to one, we randomly assign small (100 coins), medium (1,000 coins), and large (10,000 coins) sizes to hoards. This random re-sizing mimics a possible selection bias, where it may be more likely to discover hoards in specific regions and for specific time periods. We then estimate all parameters using this simulated dataset of random-sized hoard, and confront them to the known (true) parameters. As with equal-sized hoards, we recover all parameters with a high degree of accuracy. More importantly, our estimated parameters do not have any systematic tendency to estimate larger (or smaller) economic sizes for larger hoards. This means, e.g., that we are not naively inferring that regions which are wealthy today, and therefore more likely to fund archaeological research and discover ancient coin hoards, were wealthy in the past.

Estimation with young coins only. Step 5 in our formal identification proof in section E.1 states that we may need as many generations of coins as regions for identification ($T \geq N = 13$) to guarantee the uniqueness of our solution. It does not however specify how severe the problem of underidentification would be with too few generations of coins. We explore this question numerically. We simulate a dataset of coin stocks using known parameters. Unlike our estimated parameters, we choose positive minting parameters ($M_n[t] > 0, \forall n, t$) to ensure that our results are not driven by sparsity. We first draw randomly 100,000 coins from the simulated model, using *coins of all ages*. We collect them into a simulated dataset of coin hoards (as before, we normalize all hoard sizes to 1) and estimate all parameters using this simulated dataset. Our estimator recovers the true parameters with a high level of accuracy. We then draw randomly 100,000 coins from the simulated model, using *young coins only*, i.e. coins of age 1 and 2 only. We collect those young coins into a simulated dataset of coin hoards (each normalized to size 1) and estimate all parameters using this simulated dataset. Our estimator is unable to recover all true parameters. As expected from our asymptotic identification proof, our ability to recover the parameters governing bilateral trade flows is not deteriorated by the loss of information on older coins: this step in the identification is performed using only two consecutive generations of coins (see equation E.8 in step 1 of the proof in section E.1), and uses logs to control for *any* seller terms and minting parameters with two-way fixed effects. But the estimator fails at systematically disentangling seller terms from minting. A closer inspection of our (underidentified) estimates for minting reveals that while we are able to approximately recover the relative sizes of minting output within a period, we are unable to identify the levels of minting across periods. We use different colors for different periods. For each period, estimated versus true minting are approximately aligned along lines with slopes equal to one; but the levels (intercepts) of those lines differ across periods. This inability to recover the correct scale across periods is to be expected from theorem 2 in step 5 of the proof in section E.1.

Bias against old and distant coins (temporal and geographic 'home bias'). To assess the degree to which our estimates may be biased because we mistakenly ignore a bias against old and distant coins, we proceed as follows. We simulate a synthetic dataset using a (known) model where the coin loss rate is different for coins minted recently and close by ($\lambda_{\text{low}} = 0.25$) and for coins minted long ago and far away ($\lambda_{\text{high}} = 0.35$). We then estimate on this synthetic dataset the parameter of a misspecified model which *incorrectly* assumes a constant $\lambda = 0.3$ for all coins irrespective of where and when they were minted. Our misspecified model leads us to incorrectly estimate a distance elasticity larger than the true (known) elasticity: 3.13 versus 3.04. However, despite the substantial misspecification of the coin loss rate, the difference between the estimated and true distance elasticity remains small. All other parameters governing bilateral trade flows are equal or almost equal to their true (known) value.

Pro- and anti-Islamic biases (religious 'home bias'). To quantify the extent to which our estimate of the religious border effect may be biased if we mistakenly ignore a home bias against Islamic coins in non-Islamic regions, and against non-Islamic coins in Islamic regions, we perform a similar exercise. We simulate a synthetic dataset using a (known) model where the coin loss rate is systematically different for coins minted in a region with the same or a different religion. Formally, we assume that if a coin was minted in a non-Islamic (respectively Islamic) region, it is more 33% more likely to be reminted when used in an Islamic (resp. non-Islamic) region ($\lambda_{\text{high}} = 0.4$) than when used in a non-Islamic (resp. Islamic) region ($\lambda_{\text{baseline}} = 0.3$).

We use our baseline dynamic model of coin diffusion, equation (10), except for the coin loss rate which varies between λ_{high} and $\lambda_{\text{baseline}}$ depending on where and when a coin is used, to simulate a dynamic of coin stocks. From this simulated model, we draw, with replacement, 100,000 synthetic coins, which we combine into a synthetic dataset of coin hoards. We then use this synthetic dataset to estimate the parameter of our *misspecified* model, incorrectly assuming a constant $\lambda = 0.3$. Misspecifying the process for reminting, and ignoring a systematic religious bias in reminting, induces a bias in our estimates for the border effect. However, as earlier, despite introducing a large misspecification for the reminting process, the bias in our border estimates remains relatively small. Those biases are small in comparison, for instance, to the very large fluctuations in the geographic composition of Mediterranean trade we document in section 5.2. We also note that, as earlier, the other parameters governing bilateral trade (distance elasticity and political border effect) are unaffected by this religious misspecfication bias.

To conclude, we confirm that a violation of our fungibility assumption leads to biased estimates, but we also confirm that those biases are likely to be quantitatively small.

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